

CacheCraft

*When Repetition Becomes a First-Class Citizen:
A Definitive Report on Semantic Caching*

SUPPORTVECTORS AI SCHOOL

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There are only two hard things in Computer Science: cache invalidation and naming things.

— attributed to Phil Karlton

No man ever steps in the same river twice, for it is not the same river and he is not the same man.

— Heraclitus, as reported by Plutarch

This report is a sibling to RETRIEVALCRAFT, which named the semantic cache “the front door of the architecture” and gave it two pages. This volume gives the front door its own book: the geometry that makes it possible, the decision theory that makes it safe, the adversary that makes it dangerous, and the economics that decide whether to build it at all.

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Foundations: The Front Door and Its Borders

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THIS REPORT EXISTS BECAUSE A THRESHOLD I HAVE TAUGHT FOR THREE COHORTS STOPPED BEING A NUMBER. In the RAG course the semantic cache gets a full session: the lookup formalism, the two objects that carry all the intelligence, the three-tier cascade, the invalidation triple, the fine-tuned encoder. That material is sound, and I will not re-teach it here — where this report touches it, it cites it and moves on.¹ But the material is sound the way a 2023 map of a fast-growing city is sound: the streets it shows are real, and the city has since built three new districts. Between one cohort and the next, the field decided that the similarity threshold is not a scalar to be tuned but a statistical decision to be guaranteed; an adversary published a working attack on the caches we teach; and the economics that justified the whole enterprise in 2023 were quietly inverted by the model providers themselves. Three new districts. This report is the updated map — and, in the parts where the taught material leaned on assertion, the survey that should have preceded the assertion.

THE PREMISE OF CACHING IS REPETITION, and the premise holds better than we usually dare to claim in class. Measured across millions of production requests, more than seventy percent of queries arriving at a large LLM service have a semantically similar predecessor already answered.² The distribution is the usual power law: a small head

¹ The taught treatment: *Semantic Caching: Beyond String Equality*, RAG course, Session 9; and RETRIEVALCRAFT Part VI, §rc6-cache, “The Cache as the Front Door,” whose two pages are the seed of this volume.

² Yu et al., *IC-Cache / EchoLM*, arXiv:2501.12689 (2025): across production-scale traces, >70% of requests have semantically similar counterparts. Domain redundancy estimates from QVCache (arXiv:2602.02057): ~30–40% web search, 55–68% recommendation, ~70% RAG workloads.

of questions asked incessantly, a long tail asked once. Every system that answers questions for a living eventually rediscovers the same economics — *the cheapest retrieval is the one that never runs* — and builds a front door that recognizes returning questions before the expensive machinery wakes up. RETRIEVALCRAFT named the semantic cache the front door of the retrieval architecture and the cheapest judge in the whole court. This volume is what happens when the front door gets its own book.

The Object, Restated Once

THE FORMALISM TAKES FOUR LINES, BECAUSE WE BUILT IT CAREFULLY THE FIRST TIME. A semantic cache is a set of entries $\mathcal{C} = \{(e_i, r_i, t_i)\}$ — an embedding, a response, a timestamp, plus the provenance and version fields the production schema adds. A new query q is embedded by the encoder E ; the cache serves entry $i^* = \arg \max_i \text{sim}(E(q), e_i)$ if and only if $\text{sim}(E(q), e_{i^*}) > \tau$; otherwise the full pipeline runs and writes its answer back.³ And the doctrine that organizes everything downstream of the formalism, quoted verbatim from RETRIEVALCRAFT because it must recur in every treatment this house produces: “*the entire intelligence of a semantic cache lives in two objects: E and τ . Every failure you will ever debug traces to one of them.*”

That doctrine is the load-bearing wall of this report, and I want to say precisely what this report does to it, because nothing in what follows demolishes it — what follows *promotes* it. In the taught material, E is a fine-tuned encoder and τ is a scalar chosen by an offline sweep. Part II derives, properly this time, the geometry that makes the whole scheme possible — why a threshold on cosine similarity can separate signal from noise at all in a space of a thousand dimensions. Part III then shows where that geometry goes silent: the modern literature’s central empirical finding is that correct and incorrect hits *overlap* in similarity, differently for every cached entry, so no single scalar τ — however lovingly swept — can be right for all of them. The threshold becomes a per-entry statistical decision with an error budget, and the two-object doctrine survives with its second object grown from a number into a policy.

³ The formalism, the entry schema, and the three-tier production cascade (exact-hash tier, semantic tier at τ_{full} , partial-enrichment band above τ_{partial}) are taught in full in the Session 9 chapter and RETRIEVALCRAFT §rc6-cache; the GPTCache citation of record is Bang, NLP-OSS @ EMNLP 2023.

The rhyme with MEMORYCRAFT is exact and will be honored throughout: there, the read-versus-evict decision was “one function read in two directions”; here, the serve-versus-recompute decision undergoes the same promotion from scalar rule to learned policy under a gate.

What this report adds to the taught material

The Session 9 chapter answers: *what is a semantic cache, where does it sit, and how do I build one with a fine-tuned encoder and a swept threshold?* This report adds four questions the field answered after that chapter was written. *When is a threshold allowed to say yes?* — decision theory with guarantees (Parts II–III). *What exactly should a RAG system cache?* — five layers, not one (Part V). *Who is attacking the cache?* — a demonstrated adversary (the Interlude). *Does the cache still pay?* — the post-prefix-caching economics (Part VII). None of the four displaces the taught architecture; all four decide whether and how to deploy it.

A Longer Lineage Than We Taught

THE TERM “SEMANTIC CACHING” IS THIRTY YEARS OLD, and the gap between its original meaning and our current one explains, with uncomfortable precision, where all the modern problems come from. In the database literature of the mid-1990s, a semantic cache stored query *predicates* together with their results: the cache knew it held “employees with salary > 80,000,” and when a new query asked for “employees with salary > 100,000,” it could *prove* containment, answer from the cache, and ship only the remainder query to the server.⁴ Equivalence was *decidable*. A hit was a small act of logic.

The LLM era kept the name and replaced the logic with geometry: containment became cosine proximity in an embedding space. That substitution is what makes the modern cache general — no schema, no predicate language, any natural-language question — and it is also the original sin from which every chapter of this report descends. Proximity is not containment. “Basel III capital requirements” and “Basel IV capital requirements” sit a breath apart in any general embedding space and demand different answers; the 1996 cache could never confuse them, because no containment proof exists; the 2023 cache confuses them by design unless E or τ is taught otherwise. When Part VI shows the OLAP world rebuilding decidable equivalence for analytics queries — canonical intent signatures with zero spurious matches — you should hear the 1996 cache laughing softly from the back of the room. The arc of this report is, in one sentence: *we traded a theorem for a bet, the bet made the cache universal, and the field has spent three years learning to price the bet honestly.*

⁴ Dar, Franklin, Jónsson, Srivastava, Tan, *Semantic Data Caching and Replacement*, VLDB 1996. The cache reasons over predicate containment; a hit is a theorem, not a bet.

A Field Guide to the Borders

FOUR OTHER THINGS ARE CALLED “CACHING” IN AN LLM SYSTEM, and conflating them with the semantic cache is the most common confusion in both vendor copy and — I must record this honestly — one line of our own teaching. The borders are drawn by four questions: *what is the key? Is the match exact or approximate? What is reused? And what does failure look like?*

Provider prefix caching. Anthropic’s `cache_control`, OpenAI’s automatic caching, Gemini’s context caching. The key is an *identical token prefix*; the match is exact; what is reused is the provider-side KV state of that prefix; failure is impossible in the correctness sense — a miss merely costs full price. All three providers converged on roughly ninety-percent discounts for cached input tokens.⁵ Note what is *not* discounted: the model still generates every output token. Prefix caching makes computing the answer cheaper; it never avoids computing it.

KV-cache reuse research. The academic line that reuses key-value states *beyond* exact prefixes: modular segment reuse, per-chunk KV pre-computation for RAG with selective recomputation to restore cross-attention.⁶ The key is token content at the chunk level; what is reused is *computation*, not answers; failure is quality degradation from stale attention structure, which these papers spend their pages repairing.

Embedding caches. Text in, vector out, exact string key. A pure cost optimization with zero correctness risk, shipped by every serious framework. Calling this “semantic caching” — as marketing copy routinely does because embeddings are involved — is like calling a filing cabinet a librarian.

The semantic cache. Ours. The key is an embedding; the match is approximate; what is reused is a *finished answer*; and failure is a *wrong answer served with confidence at five milliseconds*. It is the only member of this family that can be wrong, and that single property generates this report: the geometry (Part II), the decision theory (Part III), the adversary (Interlude), and the economics (Part VII) are all consequences of being the one cache that bets.

Principle 0.1 (Only the semantic cache can be wrong). Prefix and KV caching change the *cost* of computing the answer; the semantic

⁵ Current parameters, August 2026: Anthropic — 5-minute and 1-hour TTLs, writes at $1.25\times/2\times$, reads at $0.1\times$, minimum prefix 512–4096 tokens by model. OpenAI — automatic at ≥ 1024 tokens, discounts 50–90% by model family, extended retention up to 24h. Gemini — implicit plus explicit `cachedContents` with storage billed per MTok-hour. Vendor documentation; mechanically verifiable pricing.

⁶ Prompt Cache, MLSys 2024 (arXiv:2311.04934); RAGCache (arXiv:2404.12457); CacheBlend, EuroSys 2025 best paper (arXiv:2405.16444); TurboRAG (EMNLP 2025); FusionRAG, SIGMOD 2026 (arXiv:2601.12904).

cache changes *whether you compute it at all*. Lossless caches fail into higher latency; the semantic cache fails into confident misinformation. Every design decision in this report — thresholds, verification, scoping, invalidation, the decision to build at all — follows from which side of that line you are on.

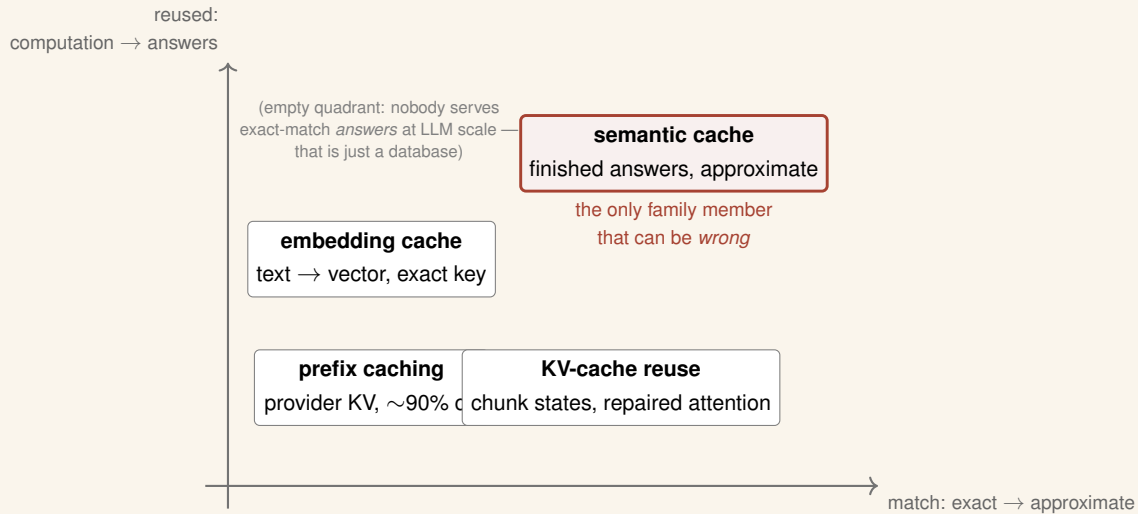


Figure 1: The four caching families on two axes. Three quadrants fail into higher latency; the fourth fails into confident misinformation. Every design decision in this report follows from which quadrant you are standing in.

AND ONE CORRECTION TO OUR OWN READING LIST, RECORDED HERE IN THE OPEN because this report’s discipline requires errors to be owned where they occurred, not smoothed over. The Session 9 lesson plan’s readings cite CacheBlend as a semantic-cache work on “knowledge fusion.” It is not: CacheBlend is a KV-cache fusion system — it stitches precomputed per-chunk KV states and selectively recomputes about fifteen percent of tokens to repair cross-attention. It belongs to the second family above, reuses computation rather than answers, and cannot serve a wrong answer. The mischaracterization is mild, but the boundary it blurs is the load-bearing one, so the lesson plan gets an erratum and this report gets a clean taxonomy.

What Changed While We Were Teaching

THE REFERENCE ARCHITECTURE WE TEACH WAS BORN IN 2023 AND DIED IN 2024, and its death is instructive. GPTCache — the modular embedding-store-threshold pipeline that anchors our Session 9 citations — published its last release in August 2024; no release has followed; its integrations bit-rotted.⁷ What killed it was not a better semantic cache.

Filed as report-inherited erratum candidate #1 in the source ledger — caught at Stage 5 of the craft pipeline, before the podium rather than after it, which is precisely the job the verification stage exists to do.

⁷ GPTCache: last PyPI release 0.1.44, 2024-08-01, none since (verified 2026-08); the LlamaIndex integration broken since 2024. “Abandoned” is an inference from two years of silence — the repository is not formally archived — but it is the inference every practitioner has drawn.

It was the provider prefix-caching wave: when the model vendors themselves began discounting repeated context by ninety percent, automatically, with zero correctness risk, the easy cost savings that 2023-era semantic caching pitched simply stopped being available to claim. The semantic cache did not become useless; it became a *specialist* — and the field reorganized around three harder questions that the easy era never forced.

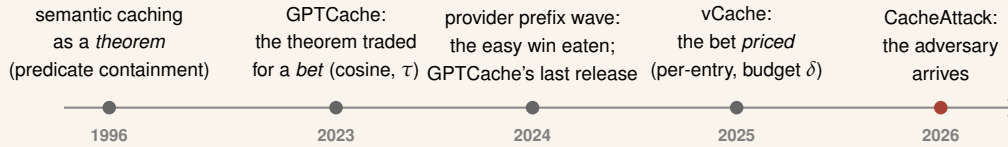


Figure 2: Thirty years in one line: a theorem, a bet, a repricing, and an adversary. The chapters of this report are this timeline, slowed down.

FIRST, THE THRESHOLD BECAME A DECISION. Berkeley’s vCache — the reference point of the modern literature — demonstrated that the similarity distributions of correct and incorrect hits overlap, entry by entry, and replaced the global scalar with per-embedding thresholds learned online under a user-specified error budget: the operator states “at most one hit in fifty may be wrong,” and the cache maintains that contract statistically, exploring when uncertain.⁸ A companion literature showed that the field had been measuring the wrong thing all along — offline ranking metrics do not predict threshold usability — and supplied the calibration methodology Part III teaches.

⁸ Schroeder et al., *vCache: Verified Semantic Prompt Caching*, arXiv:2502.03771, ICLR 2026. Up to $12.5\times$ the hit rate at $26\times$ lower error than static thresholds. Successors: MVR-cache (ICML 2026) adds multi-vector matching under the same guarantee; Krites (arXiv:2602.13165) verifies the near-miss band with an asynchronous judge.

SECOND, THE ADVERSARY ARRIVED. In January 2026 a key-collision attack was published against exactly the caches we teach: adversarial prompts, semantically unrelated to a victim query but colliding with it in embedding space, hijacked cached answers — and cached *tool invocations* — at rates between seventy-seven and ninety-one percent, against reference stacks including the major clouds’ semantic-cache architectures.⁹ The uncomfortable theorem inside the attack: hit rate and collision resistance are the same quantity with opposite signs. A shared, unscoped semantic cache is no longer a design smell; it is a published exploit target. Our taught material’s five dangers said “privacy leak” in one line; the Interlude owes you the full anatomy.

⁹ Zhang et al., *From Similarity to Vulnerability: Key Collision Attacks on Semantic Caches*, arXiv:2601.23088, ICML 2026. The paper’s Lemma 3.1 proves the tradeoff is structural: the locality that produces hit rate is the same locality that produces collisions.

THIRD, THE ECONOMICS INVERTED. With prefix caching capturing the repeated-context savings automatically, the semantic cache’s surviving value proposition narrowed to the two things prefix caching cannot touch: the *output* tokens it never generates, and the seconds of latency it never spends. Those are real — output tokens dominate many bills,

and sub-hundred-millisecond answers are a product category of their own — but they pay only on genuinely repetitive, non-personalized, non-time-sensitive traffic, which honest measurements put at fifteen to forty-five percent of real workloads depending on shape, a far cry from the ninety-five-percent hit rates of vendor calculators.¹⁰ Part VII does this arithmetic honestly, including the configurations where the correct engineering decision is the one our taught viability gate already implies: do not build the cache.

The Shape of This Report

THE MAP, THEN, PART BY PART. Part II is the geometry — the concentration-of-measure treatment this house has taught in three courses and never once connected to the cache’s threshold, done in depth, with the polar cap, the equatorial waste, and the polar bears who live where almost nothing else can. Part III is the decision theory: what the cap cannot decide, the threshold-as-policy must. Part IV is the modern lookup stack — cache-purpose embedders, calibration-aware training, multi-turn keys — where the taught fine-tuning pipeline gets its 2026 refresh. Part V asks what, exactly, a RAG system should cache, and answers with five layers where we taught one. Part VI takes the cache to agents, where cosine similarity turns out to be the wrong equivalence relation entirely. The Interlude arms the adversary. Part VII runs the books. Part VIII walks the industry as it actually is — previews, pivots, abandonments, and the two vendors whose own documentation warns against their own product. Part IX distills the patterns and closes the rhymes with the sibling crafts. The appendix holds the glossary and a reading list ordered by dependency, ending where the field currently stands: at a verified threshold and a demonstrated attack.

Step back and see the shape of the argument before we begin. A cache is a bet that the past will repeat. The geometry of high dimensions makes the bet *possible*; the decision theory makes it *honest*; the adversary makes it *dangerous*; the economics decide whether it is *worth placing at all*. Four verdicts on one bet — and a threshold that grew up, somewhere between two cohorts, from a number into a policy.

¹⁰ Practitioner hit-rate ladder across 2026 sources: FAQ/support 40–60%; classification 50–70%; RAG Q&A 15–25%; open-ended chat 10–20%; agentic tool traffic 5–15%. The “95% hit rate” of vendor copy is precision-given-a-hit laundered into recall — a conflation Part VII dissects with the ledger discipline of RETRIEVALCRAFT Part IX.

Yes, the bears. If you have sat in any of my embedding lectures you have met them; Part II finally gives them a formal habitat.

The Geometry: The Cap and the Equator

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THIS HOUSE HAS TAUGHT CONCENTRATION OF MEASURE THREE TIMES, in three courses, at three altitudes — and never once walked it to the place where it earns its keep in this report: the moment a semantic cache says *yes*. The LLM course meets it phenomenologically in week two: sample random vectors, watch the cosines huddle around zero, learn that “in high dimensions, randomness is not noisy; it is predictable.” The fine-tuning course states it formally, in a margin, while teaching encoder fine-tuning: Lévy’s inequality on the unit sphere, the Goldilocks band of dimensions, the north pole and the equatorial waste. RETRIEVALCRAFT distills it into the aperture doctrine: nothing wanders into the polar cap by chance. Three treatments, one theorem, and a threshold — τ , the second of our two sacred objects — that has been living off this theorem’s credit without ever being introduced to it. This part performs the introduction, in depth, because the entire legitimacy of thresholded similarity rests here, and because the exact place where the theorem goes *quiet* is the exact place Part III must take over.

This is also the part where the polar bears, who have padded through my embedding lectures for years, finally receive a formal habitat. Patience; they arrive in §.

The Strangeness, Experienced First

RUN THE EXPERIMENT BEFORE NAMING THE PHENOMENON — the house rule. Sample two independent random unit vectors in $d = 3$ dimensions and record their cosine similarity; repeat ten thousand times; draw the histogram. It is broad and flat-shouldered: in three dimensions, random directions are frequently oblique, sometimes nearly aligned, occasionally opposed. Now set $d = 768$ — the working dimension of a modern sentence encoder — and repeat. The histogram collapses into a needle centered at zero: virtually every pair of random directions is virtually orthogonal. At $d = 1536$ the needle is sharper still, its width shrinking like $1/\sqrt{d}$.

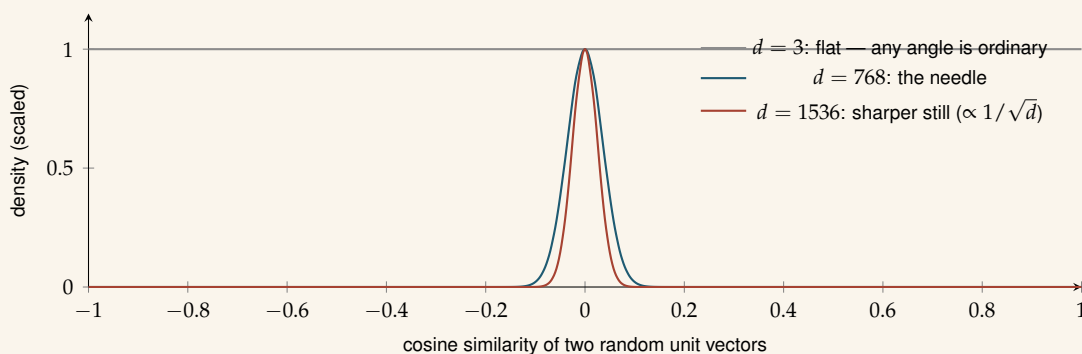


Figure 3: The week-two experiment, redrawn: the exact density of the cosine between two random unit directions, $\propto (1 - c^2)^{(d-3)/2}$. In three dimensions every angle is ordinary; at working dimensions randomness condenses onto orthogonality. Whatever appears far from the needle was put there.

Three sentences of interpretation, imported from the LLM course because they cannot be improved: in high dimensions, randomness is not noisy; it is *predictable*. Almost every random vector is nearly orthogonal to every other. Orthogonality is not a special relationship — it is the default. The same collapse governs lengths: sample the coordinates of a vector from independent standard Gaussians, and its norm lands within a constant of $\sqrt{d}$ with overwhelming probability — the *Gaussian annulus* phenomenon. High-dimensional randomness does not spread out; it *condenses* onto thin shells and thin bands, and all of high-dimensional geometry’s famous strangeness — spheres with their volume at the surface, cubes with their volume in the corners, oranges that are all peel — is this one condensation wearing different costumes.¹¹

¹¹ Blum, Hopcroft, Kannan, *Foundations of Data Science*, Cambridge University Press, 2020, Ch. 2 — the cleanest elementary treatment of the annulus theorem and surface-concentration phenomena, and the canonical reference this house assigns.

The taught corpus already owns the two intuition anchors, and this report gladly inherits them. *Pennies in a jar*: a transformer accumulates contributions into its residual stream the way a jar accumulates pennies, so longer texts grow longer vectors — magnitude measures accumula-

tion, not meaning, which is why the cache normalizes every embedding onto the unit sphere before comparing directions and why the Session 9 chapter calls normalization “the invisible load-bearing wall.” And the *Bay Area versus Wyoming* rule: nearest does not mean near. In an empty stretch of Wyoming your nearest neighbor may be forty miles away; in high dimensions, *everything* is Wyoming, and the argmax of a similarity search is a statement about ranking, never about closeness — which is precisely why RETRIEVALCRAFT’s sternest principle forbids running a k -nearest-neighbor search without a threshold.

The Theorems, Stated Honestly

NOW THE MACHINERY, WITH THE CARE THE MARGIN NOTES NEVER HAD ROOM FOR. Fix the unit sphere $S^{d-1} \subset \mathbb{R}^d$ with its uniform probability measure μ . The master fact is *Lévy’s concentration inequality*: for any function $f : S^{d-1} \rightarrow \mathbb{R}$ that is 1-Lipschitz (it cannot amplify distances), the measure of points where f strays from its central value decays exponentially in the dimension,

$$\Pr[|f(v) - \mathbb{E}[f]| > \varepsilon] \leq 2 \exp\left(-\frac{(d-1)\varepsilon^2}{2}\right).$$

Read it aloud: on a high-dimensional sphere, *every* well-behaved function is nearly constant almost everywhere. The sphere leaves its functions no room to vary.¹²

The cache cares about one particular Lipschitz function. Fix a query direction u (call it the north pole) and let $f(v) = \langle u, v \rangle$ — the cosine similarity itself, since everything is normalized. It is 1-Lipschitz, and by symmetry its mean over a random v is zero. Lévy immediately hands us the fact the whole cache stands on:

$$\Pr[|\langle u, v \rangle| > \varepsilon] \leq 2 \exp\left(-\frac{(d-1)\varepsilon^2}{2}\right) \approx 2 \exp\left(-\frac{d\varepsilon^2}{2}\right),$$

the right-hand form being the large- d asymptotic one meets in direct cap-measure derivations. *A random direction almost never has non-trivial cosine similarity with your query.*

¹² Lévy’s inequality descends from the isoperimetric inequality on the sphere; Ledoux, *The Concentration of Measure Phenomenon*, AMS 2001, is the definitive account, and Vershynin, *High-Dimensional Probability*, CUP 2018, the modern teaching text. A hygiene note this corpus now needs on record: the constants and the centering (mean versus median) vary across authors and normalizations; the exponent’s *shape* — dimension times ε^2 — is the invariant content.

One theorem, two exponents — a reconciliation

The fine-tuning course’s margin states the bound with $(d-1)$; RETRIEVALCRAFT Part II states it with d . Neither is wrong, and they are now formally reconciled: $(d-1)$ is Lévy’s sphere-exact

exponent; d is the asymptotic form from the direct estimate of cap measure, and at $d = 768$ the two *exponents* differ by less than two parts in a thousand (the bound values differ by the modest factor $e^{\varepsilon^2/2}$). This report adopts the exact $(d - 1)$ form when stating the theorem and the round d form when doing arithmetic — and the corpus’s two earlier statements should be read as the same claim at two levels of pedantry. What *was* wrong, and is hereby retired, is the week-9 errata’s attempted paraphrase, which conflated two true statements — “the sphere’s measure huddles at the equator” and “the cap around the query is exponentially tiny” — into one muddled one. They are the same fact seen from the pole and from the band: the equator is where measure lives *because* every polar cap is starved of it.

HOW STARVED? THIS IS WORTH A NUMBER, because the numbers are the argument. The spherical cap of all directions with cosine at least ε from the pole has measure at most $e^{-d\varepsilon^2/2}$, and more precisely of order $(\sin \theta)^{d-1}$ for angular radius θ — exponentially small in dimension, doubly so as the cap tightens.¹³ At the working dimension $d = 768$:

cosine ε	0.10	0.20	0.50	0.85
chance-hit bound $e^{-d\varepsilon^2/2}$	2.1×10^{-2}	2.1×10^{-7}	2×10^{-42}	3×10^{-121}

¹³ The cap-measure bound $\mu(\{v : \langle u, v \rangle \geq \varepsilon\}) \leq e^{-d\varepsilon^2/2}$ is classical; see Ball, *An Elementary Introduction to Modern Convex Geometry*, MSRI 1997, Lemma 2.2.

The left-hand column carries its own lesson: at $\varepsilon = 0.1$ the bound is only a few percent. Low thresholds are not “looser” — they are a different regime, where chance collisions are physically real.

The taught warning against wide apertures ($\tau = 0.70$, the wide cone of the Session 9 figure) is quantitative, not aesthetic.

History: *Statistical mechanics got here first: the annulus is the microcanonical energy shell. Boltzmann’s insight — that equilibrium is simply the overwhelming majority of microstates — is concentration of measure, sixty years before Lévy named it. The second law of thermodynamics is a theorem about where high-dimensional volume lives.*

¹⁴ Johnson, Lindenstrauss, *Extensions of Lipschitz mappings into a Hilbert space*, *Contemp. Math.* 26, 1984. The compression figure: MeanCache, arXiv:2403.02694.

Stare at the right-hand columns. At the thresholds we actually teach — 0.85 and above, the folklore band around 0.92 — the probability that an *unrelated* text lands inside the cap by chance is not small; it is cosmologically zero, a number with no physical process behind it. Whatever shows up at cosine 0.85 from your query was *put there*.

Two more classical facts complete the toolkit, each with one cache-shaped consequence. The *Gaussian annulus theorem*: a d -dimensional standard Gaussian vector has norm $\sqrt{d} + O(1)$ with high probability — deviations of width t carrying failure probability $e^{-\Omega(t^2)}$ — so raw embedding magnitudes are dimension artifacts, and comparing unnormalized vectors mixes accumulation with meaning; the pennies-in-a-jar doctrine is this theorem in street clothes. And *Johnson–Lindenstrauss*: any n points can be projected into $O(\varepsilon^{-2} \log n)$ dimensions while distorting all pairwise distances by at most $1 \pm \varepsilon$ — which is why a 768-dimensional space is not merely adequate but *lavish* for a cache of a few million entries, and why the embedding-compression results in the caching literature (an eighty-three percent storage reduction with negligible matching loss) are not surprising but predicted.¹⁴

The Polar Bears

NOW THE PICTURE THAT CARRIES EVERYTHING, DRAWN SLOWLY. Fix your query and call it the north pole. The theorem has just told us where everything *random* lives: in the vast equatorial band, ninety degrees from you, give or take a hair's width that shrinks like $1/\sqrt{d}$. The equator is the land of the irrelevant — every unrelated document, every off-topic query, every stranger. It is, overwhelmingly, *all* of the sphere. And the polar cap around your query is a desert of measure: exponentially tiny, cosmologically improbable, a place chance simply cannot reach at working dimensions and taught thresholds.

So consider what it means to find something living there. The polar cap is like the high Arctic: almost nothing can live there, and *that is exactly why everything you meet there is significant*. A polar bear does not wander to the pole. It is not there by accident, blown off course from the tropics. It is there because it is *adapted* to be there — because a long optimization shaped it for precisely that habitat. When you meet a polar bear in the cap, you are not meeting a coincidence; you are meeting the end product of a process that pulled it poleward. In embedding space the process has a name: *training*. Contrastive learning is the evolutionary pressure of this landscape — it breeds cap-dwellers, pulling semantically kin texts into each other's polar caps and banishing the unrelated to the equator where measure says they belong. RETRIEVALCRAFT said it in one line: *signal survives concentration; noise is annihilated by it*. The bears are the survivors.

Principle 0.2 (The cap population is curated, not sampled). At working dimensions and taught thresholds, chance contributes nothing to a polar cap. Every occupant was placed there by learning — so a cache hit above threshold is never a fluke, and *whatever* is wrong with a wrong hit, it is not bad luck. This cuts both ways: it is the guarantee that makes thresholded caching sound against noise, and it is the warning that every failure is systematic — a property of *E*'s training, not of the dice.

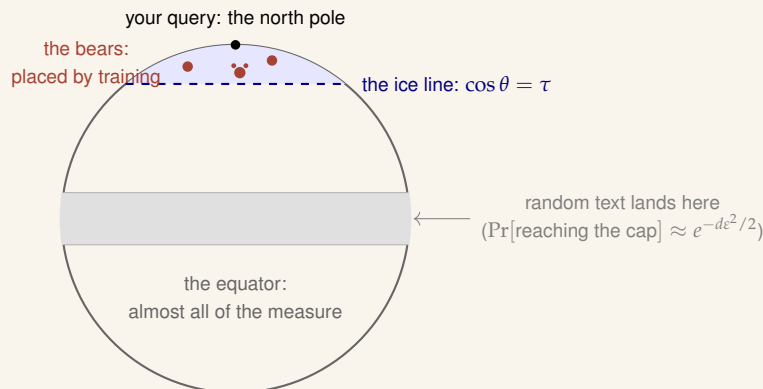


Figure 4: The habitat, drawn honestly. The equatorial band holds essentially all of the sphere’s measure — the land of the irrelevant. The polar cap above the ice line τ is starved of chance measure; its population consists exclusively of polar bears, occupants adapted to it by contrastive training. One of them, on close inspection, has ears.

The aperture doctrine — tighten the cone for precision, widen it for recall — is unchanged and remains the right first lecture.

What this section adds is the population the aperture is actually selecting over: not signal-versus-noise, which the geometry settled, but bear-versus-bear, which it did not.

This is the deep reason the semantic cache works at all, and it deserves to be said as directly in the lecture as it is said here: the threshold τ is not filtering noise from signal by degrees. At $\tau = 0.85$ in 768 dimensions there *is no noise to filter* — concentration already annihilated it. The threshold’s true job is different, and subtler: it adjudicates *among the bears*. Hold that thought; it is the hinge of this report.

Where the Theorem Goes Quiet

EVERY THEOREM EARNS TRUST BY ADMITTING ITS JURISDICTION, and concentration’s jurisdiction ends precisely where the cache’s troubles begin. Three honest limits, in ascending order of consequence.

FIRST: REAL EMBEDDINGS ARE NOT UNIFORM. The theorems above describe *random* directions; trained encoders emit nothing of the sort. Empirically, the embeddings of contextual models occupy a narrow cone — *anisotropy* — so that two texts chosen at random commonly score cosine 0.2 to 0.4 under popular sentence encoders, higher still under some (certain commercial embedders sit near 0.7; word-level measurements exceed 0.9 in upper GPT-2 layers) — never the theorem’s zero.¹⁵ The equator, in other words, is not where the theorem drew it; the whole population is shifted toward a common pole, the effective dimension is lower than the nominal one, and every cosine number in production is measured against this shifted background. This is why the folklore thresholds (0.85, the 0.92 sweet spot) are *dataset-relative folklore* rather than geometry: they are statements about a particular encoder’s anisotropy meeting a particular workload, and they transfer between systems exactly as well as anisotropy does, which is to say

¹⁵ Ethayarajh, *How Contextual are Contextualized Word Representations?*, EMNLP 2019 — the canonical anisotropy measurement, whose word-level numbers run even higher than the sentence-encoder folklore. The Session-9 corpus already owns a figure for this (fig-isotropy-anisotropy); it now has its formal role.

poorly. It is also one honest reason the taught fine-tuning pipeline works: contrastive fine-tuning with hard negatives spreads the cone, restoring some of the isotropy that the theorem assumed and the pretrained encoder lacks.

SECOND: THE THEOREM SEPARATES THE CAP FROM THE EQUATOR — NEVER BEAR FROM BEAR. Concentration is a statement about the overwhelming majority of the sphere, the random bulk. About the interior structure of an exponentially small cap — the fine geometry of the few dozen learned neighbors of your query — it says exactly nothing. Basel III and Basel IV are both bears. The paraphrase of your question and its treacherous near-miss (“*stressed* netting-set exposure” against unstressed; the left arm against the right) are both bears. Type 1 and Type 2 diabetes: bears, nose to nose in the same cap, demanding different answers. Within the cap, similarity differences of a few hundredths separate equivalence from error, and *those* differences are governed not by any universal theorem but by the accidents of *E*’s training data — entry by entry, neighborhood by neighborhood. This is the geometric face of the empirical fact on which the modern literature turns: the similarity distributions of correct and incorrect hits *overlap*, and overlap differently for every cached entry.¹⁶ Concentration gets you from *random* to *relevant* with a guarantee; nothing gets you from relevant to *equivalent* except a decision made under uncertainty, with evidence, per entry. That is Part III.

THIRD: A CURATED CAP CAN BE GAMED. The same locality that lets training place bears in your cap lets an *adversary* craft one — a prompt semantically unrelated to your query, gradient-tuned to land inside your cap wearing a bear suit. The 2026 collision attacks are exactly this, and their core lemma is the dark twin of this part’s principle: the locality that produces hit rate *is* the locality that produces collisions; you cannot buy one without selling the other. The cap’s guarantee — nothing arrives by chance — survives; the comfortable reading of it — everything here is friendly — does not.

The theorem’s jurisdiction, in three lines

Concentration of measure guarantees: no chance hits at taught thresholds; every occupant of the cap was placed there. It does not adjudicate among occupants (paraphrase versus hard negative), does not fix thresholds across encoders (anisotropy moves the equator), and does not authenticate occupants (the bear suit).

¹⁶ The per-entry overlap measurement is vCache’s founding observation (arXiv:2502.03771, ICLR 2026) — Part III’s subject. The geometry of this section is why the observation should not have surprised us.

Interlude, *The Adversary at the Door*. The bear suit gets a full anatomy there; here it is enough to note that the geometry which guarantees hits-are-caused does not certify *who* caused them.

One guarantee, three silences — and Parts III, IV, and the Interlude are the three silences, answered.

What the Geometry Buys the Cache

COROLLARIES, THEN, THAT REACH THE KEYBOARD. The taught error budget $\text{FPR}_{\text{total}} = \text{FPR}_{\text{embedder}} + \text{FPR}_{\text{ANN}}$ can now be read with the right physics. The ANN term is search approximation — HNSW or IVF occasionally missing the true neighbor — an engineering tax, tunable to a tenth of a percent. The chance-collision term that a newcomer would expect to dominate is, by the table above, absent entirely: it is not in the budget because at taught thresholds it is smaller than cosmic-ray error. Which means the embedder term — within-cap confusion, bear against bear — is not merely the largest term of the false-hit budget. It is, to within cosmic noise, the *only* term. Every false hit your cache will ever serve is a failure of learned local geometry, and the instruments that address it are exactly the ones the next two parts teach: better local geometry (fine-tuning, Part IV) and per-entry decisions (thresholds-as-policy, Part III).

The *Goldilocks band* of dimensions — 768 to 1536, taught in the fine-tuning course — also falls out with its reasons attached. Below the band, concentration is too weak: the needle histogram widens, caps at useful angular radii carry real chance-measure, and random collisions enter the false-hit budget for the first time. Above the band, the theorem only strengthens — but anisotropy means the *effective* dimension saturates while storage, embedding latency, and index cost grow linearly, and JL says the extra room was never needed. The band is where the guarantee is already cosmological and the invoice is still reasonable.

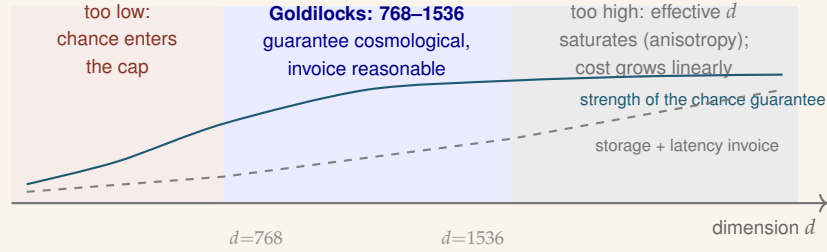


Figure 5: The Goldilocks band, with its reasons attached: below it, concentration is too weak and random collisions become physical; above it, the guarantee only strengthens while anisotropy caps the effective dimension and the invoice keeps growing.

And the taught *viability gate* — sample a hundred queries; if fewer than one in ten finds a neighbor at 0.85, do not build the cache — is revealed as a geometry test in disguise. It asks: does this workload’s query distribution actually form caps? A workload of singletons is all equator — every query a stranger to every other — and no threshold, however learned, can conjure bears onto empty ice. The gate is the cheapest possible measurement of whether repetition, the premise of Part I, holds *in embedding space*, where the cache must cash it.

STEP BACK AND SEE THE SHAPE. One theorem — measure condenses; caps starve; functions flatten — explains why the cache’s central act, thresholding a cosine, is sound at all: the equatorial world can never reach the cap, so a hit is always a caused event. The same theorem, honestly bounded, hands the report its agenda: it cannot tell bears apart, it cannot port thresholds across anisotropic encoders, and it cannot check for bear suits. The threshold we taught as a scalar was spending the theorem’s guarantee on a problem — chance — that the theorem had already solved, while the real problem — judgment among the curated — went unpriced. Pricing it is a decision-theory problem, it was solved between our cohorts, and Part III prices it — after one more chapter of geometry, because the sphere has four more stories to tell, and each of them, too, ends at a cache design decision.

The Geometry, Continued: Crowds, Hubs, and the Capacity of the Sphere

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THE PREVIOUS CHAPTER TOLD THE STORY I HAVE TAUGHT FOR YEARS, repaired and finally connected to the threshold. This chapter tells the four stories I had not taught — the adjacent geometry that the concentration lens, pointed only at the cap and the equator, does not see. Each story ends, as everything in this report must, at a cache design decision: one story sharpens the doubt about whether nearest-neighbor search means anything at all (and then rescues it), one describes what fine-tuning *actually* does to the sphere, one reveals that the sphere’s citizens are radically unequal, and one counts the sphere’s addresses and finds the number absurd. Together they finish the promotion of the aperture doctrine that the last chapter began.

The Meaningfulness Critique, and Its Rescue

THERE IS A FAMOUS THEOREM THAT APPEARS, AT FIRST READING, TO DEMOLISH THIS ENTIRE REPORT. Under broad conditions, as dimension grows, the distance from a query to its *nearest* neighbor and to its *farthest* converge: the relative contrast $(D_{\max} - D_{\min})/D_{\min}$ collapses to zero,

¹⁷ Beyer, Goldstein, Ramakrishnan, Shaft, *When Is “Nearest Neighbor” Meaningful?*, ICDT 1999. The companion result on metric choice: Aggarwal, Hinneburg, Keim, ICDT 2001 — higher L_p norms concentrate worse; fractional norms resist longest.

¹⁸ Durrant, Kabán, *When is “nearest neighbor” meaningful: A converse theorem and implications*, J. Complexity 2009 — the rescue clause. Estimators of intrinsic dimension: Levina–Bickel (NeurIPS 2004, MLE from neighbor distances); Facco et al.’s TwoNN (Sci. Reports 2017, from the ratio of the two nearest).

¹⁹ Intrinsic dimension of transformer representations: Valeriani et al. and Tulchinskii et al., NeurIPS 2023; for contextual LM embeddings and training dynamics, Bruno et al., *Less is More: Local Intrinsic Dimensions of Contextual Language Models*, NeurIPS 2025 (arXiv:2506.01034). The consistent finding: $ID \ll d$.

²⁰ LID’s extreme-value-theoretic foundation: Houle, SISAP 2017; estimation at scale: Amsaleg et al., KDD 2015. The adversarial connection: Ma et al., *Characterizing Adversarial Subspaces Using Local Intrinsic Dimensionality*, ICLR 2018 — high LID flags adversarial neighborhoods.

The Interlude’s bear suit acquires a geometric signature: a collision-crafted prompt sits in a neighborhood whose LID is measurably anomalous. A cache that logs LID per entry has a cheap adversarial tripwire nobody currently ships — capstone-sized, again.

and “nearest neighbor” asymptotically names nobody in particular.¹⁷ If that conclusion applied to our embeddings, the semantic cache would be an elaborate way of serving arbitrary answers. It does not apply, and the reason it does not is the single most load-bearing fact about real embedding spaces: *the theorem’s conditions require the data to genuinely fill the ambient dimension*, and real embeddings never do. There is a converse theorem making this precise — data with low *intrinsic* dimension escapes the collapse; relative contrast survives exactly when the data lives on a structure much smaller than its container.¹⁸

And measured, the escape clause holds with room to spare: language-model embeddings exhibit intrinsic dimensions of a few dozen — one to two orders of magnitude below their ambient 768 or 1536 — a manifold of meaning folded into a vast empty container.¹⁹ Two consequences for us. First, the honest resolution of a tension the previous chapter left implicit: the concentration guarantees (chance annihilated) and the meaningfulness critique (contrast annihilated) are *both* statements about high-dimensional randomness — and the embedding manifold, being low-dimensional and structured, inherits the first (against the random equatorial world) while escaping the second (within the learned manifold, contrast is real). The cache lives in the best quadrant: protected by the ambient dimension, discriminating within the intrinsic one. Second, a diagnostic with teeth: intrinsic dimension is *local* — it varies across the manifold — and **local intrinsic dimensionality** (LID) around a point predicts exactly the things a cache operator cares about: how stable that point’s neighborhood is, how trustworthy its nearest-neighbor relations are — and, remarkably, how exposed it is to adversarial approach, since adversarial regions are precisely regions of anomalously high LID.²⁰

The Metropolis Effect: What Fine-Tuning Actually Does

A MISCONCEPTION HIDES IN THE WAY WE USUALLY NARRATE FINE-TUNING, AND I WANT TO CORRECT IT IN PRINT — because I have caught myself implying it at the board. The narration goes: the pretrained encoder is anisotropic, a narrow cone; fine-tuning restores isotropy; therefore fine-tuning *spreads the points out*. The first two clauses are true. The third does not follow, and is false in the way that matters most. Reducing anisotropy does not mean everything becomes more diffuse. What domain fine-tuning actually produces — and what the theory says its objective *must* produce — is a sphere that is globally spread and locally

condensed: broad occupancy of the surface, punctuated by pockets of intense concentration. *The sphere does not become a uniform gas; it urbanizes.* Call it the **metropolis effect**: tight, dense downtowns of near-equivalent meaning — every paraphrase of “what is the LCR definition” packed within a whisker of each other — separated by open country that the domain’s semantics genuinely does not inhabit.

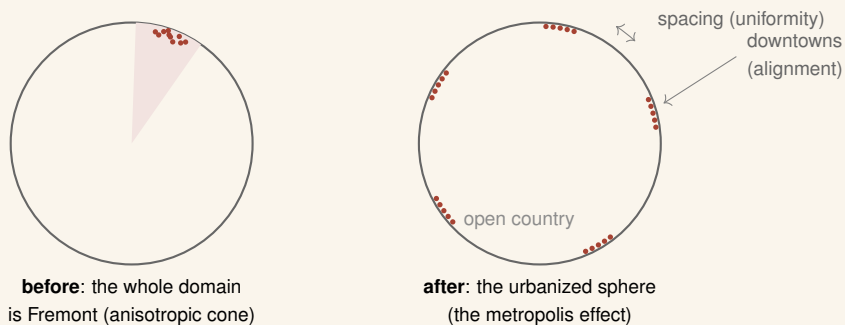


Figure 6: What domain fine-tuning actually does. Left: under a general encoder the domain is compressed into a narrow cone — one crowded pixel of the globe. Right: contrastive fine-tuning does not diffuse the points; it urbanizes the sphere — alignment packs near-equivalent meanings into tight downtowns, uniformity spreads the downtowns apart, and the temperature is the zoning law that sets their density. Threshold difficulty concentrates exactly where the traffic does: in the metropolises.

The theory behind this is the most elegant result in contrastive learning, and it deserves a place beside InfoNCE in the taught pipeline. The contrastive objective provably decomposes into exactly two asymptotic forces: **alignment** — positive pairs pulled to the same point — and **uniformity** — the feature distribution pushed toward uniform on the sphere, formalized as minimizing a Gaussian-kernel energy over pairs.²¹ Alignment builds the metropolises; uniformity spreads the metropolises apart — it acts on the distribution of mass, and its optimum is perfectly consistent with tight local clusters whose *centers* tile the sphere. And the zoning law between them is the temperature: the previous chapter promoted the temperature–threshold coupling to a principle, and the theory now says why — temperature directly governs the uniformity–tolerance tradeoff, low σ punishing near neighbors into tighter, more separated downtowns, high σ tolerating semantic sprawl.²²

THE CACHE CONSEQUENCES ARE IMMEDIATE, AND THEY CLOSE A LOOP THE DECISION CHAPTER OPENED. The metropolises are *cache country*: dense repetition, populated caps, high hit rates — and simultaneously the districts where hard negatives crowd closest, where Basel III presses against Basel IV in the same downtown block. The open country is the long tail: no neighbors, no hits, no danger. Which means threshold difficulty is not distributed uniformly over the sphere — it is concentrated precisely where the traffic is, in the downtowns, neighborhood

²¹ Wang, Isola, *Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere*, ICML 2020. The uniformity term is what fights the pretrained cone of Gao et al.’s *representation degeneration* (ICLR 2019); alignment is what builds the downtowns.

Deep Dive: The uniformity term is physics verbatim: minimizing a pairwise Gaussian-kernel energy over the sphere is the Thomson problem — point charges repelling into maximal spread. Which makes the metropolis effect cosmic structure formation in miniature: alignment plays gravity, uniformity plays expansion pressure, and the fine-tuned sphere ends up looking like the universe does — clusters, filaments, and voids. The training temperature is not a metaphor for temperature; it is the same β^{-1} in the same Boltzmann factor.

History: And the name pays a double courtesy: the sphere urbanizes, and Nicholas Metropolis — whose 1953 Los Alamos algorithm taught statistical mechanics to sample exactly such energy landscapes — smiles from the margin.

²² Wang (Feng), Liu (Huaping), *Understanding the Behaviour of Contrastive Loss*, CVPR 2021 — the temperature analysis. (A different pair of Wangs from the ICML paper; the coincidence confuses citations everywhere.)

The taught Fremont-on-the-globe story now has its full second act. Under the general encoder, the whole domain is Fremont — one crowded pixel. Domain fine-tuning promotes Fremont to the entire globe... whereupon it promptly develops its own San Francisco, its own suburbs, and its own Wyoming. The domain uses more of the surface; the surface is anything but uniform.

²³ Radovanović, Nanopoulos, Ivanović, *Hubs in Space: Popular Nearest Neighbors in High-Dimensional Data*, JMLR 11:2487–2531, 2010 — the canonical treatment. Hubness measured and reduced in modern sentence-embedding spaces: Nielsen, Sandholt, Hansen, NLDL 2024 (arXiv:2311.18364) — ~75% hubness reduction, ~9% error reduction on SBERT-class spaces.

History: *The network scientists will feel at home: hubness is the rich-get-richer signature of Barabási's scale-free networks, emerging here not from growth dynamics but from pure geometry — the same small-world structure the taught HNSW section already leans on for search now reappears as a pathology of matching.*

nearest-neighbor relations

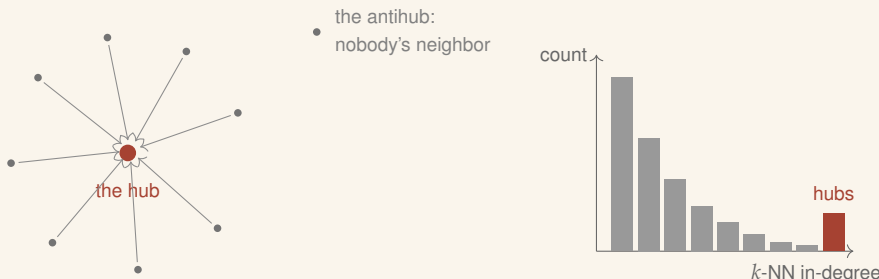


Figure 7: Neighborhood is not a democracy. As dimension grows, the in-degree of the k -NN relation becomes violently right-skewed: a small aristocracy of hubs appears in a huge fraction of neighbor lists while antihubs vanish from all of them. A hub *cache entry* is a false-hit magnet that reads, on the dashboard, as the cache's greatest success — audit the serve-count tail, and rescale with CSLS.

by neighborhood. That is the geometric restatement of the overlap measurement that motivated per-entry thresholds: *a global τ is a single zoning rule for Manhattan and Wyoming*. And it hands the fine-tuning pipeline a sharper objective than “restore isotropy”: what the cache wants from E is well-separated metropolises with maximal spacing between downtowns whose answers differ — which is exactly what curriculum-mined hard negatives do, block by block.

The Unequal Citizens: Hubs

THE THIRD STORY IS A GENUINE STRANGENESS OF HIGH-DIMENSIONAL NEIGHBORHOODS, known to the similarity-search literature for fifteen years and almost never mentioned in ours: nearest-neighborhood is not a democracy. As dimension grows, the distribution of “how often is this point somebody's nearest neighbor” becomes violently right-skewed: a small aristocracy of **hubs** appears in the neighbor lists of a huge fraction of all queries, while a matching class of *antihubs* is nobody's neighbor at all — and this is not an artifact of bad data but an intrinsic consequence of concentration itself, points slightly closer to the data mean becoming slightly closer to *everything*.²³

For a semantic cache this is not a curiosity; it is a named failure mode this corpus did not have a name for. A hub *cache entry* is a false-hit magnet: it sits in the retrieval path of queries from all over the sphere, wins nearest-neighbor contests it has no business entering, and — because it keeps getting served — looks, on the hit-rate dashboard, like the cache's most successful citizen. The taught near-miss mining would eventually surface its victims; the hubness literature says you can find the culprit directly: audit the *serve-count distribution* per entry, and treat extreme right-tail entries as geometric suspects, not successes. Better still, the correction is a drop-in: **CSLS** — penalize a candidate's

similarity by the average similarity of its own neighborhood, so that promiscuous points pay for their popularity — was built for exactly this pathology in cross-lingual retrieval and transfers verbatim to cache matching.²⁴

AND THE STORY HAS JUST ACQUIRED AN ADVERSARY, in a paper that reads like this report’s Interlude wearing hubness clothes: *adversarial hubness* — an attacker crafts a single point engineered to be a hub, a universal nearest-neighbor retrieved by millions of unrelated queries. Against embedding-based retrieval this is a poisoning amplifier of a different magnitude than the one-victim collision attack: one implanted entry, served to everyone.²⁵ The defenses rhyme with the Interlude’s — scoping shrinks the audience a hub can reach, verification catches the nonsense it serves, and now CSLS and serve-count audits join the toolkit as the hub-specific tripwires. The bear suit was one impostor in one cap; the adversarial hub is an impostor who becomes everyone’s neighbor. Same geometry, larger blast radius, same conclusion: locality is power, and power draws attackers.

The Capacity of the Sphere

THE LAST STORY ANSWERS A QUESTION SO BASIC IT IS RARELY ASKED: HOW MANY DISTINGUISHABLE MEANINGS FIT ON THE SPHERE? If the answer were “about d ” — one per dimension — then 768 dimensions would be a crowded filing cabinet and near-identical meanings would be forced to share addresses. The truth is stranger: insist only that directions be *nearly* orthogonal — pairwise cosine below some small ε — and the sphere holds not d of them but $e^{c\varepsilon^2d}$: *exponentially* many in the dimension. Relax orthogonality by a whisker and the address book explodes.²⁶ At $d = 768$ the number is astronomical — and this, precisely stated, is the correct version of an intuition worth engraving: *it is the sphere’s address book, not its dimension, that is exponential*. There is room for every question every user will ever ask, each with its own address, even when their semantic content is nearly identical. Points sit on top of each other only if the encoder *puts* them there.

²⁴ Conneau, Lample, Ranzato, Denoyer, Jégou, *Word Translation Without Parallel Data*, ICLR 2018 — CSLS. The elder sibling: Schnitzer et al.’s Mutual Proximity, JMLR 2012. Both are similarity *rescalings*; both compose with everything in the decision chapter.

Sibling rhyme, unexpected: CSLS is a *per-entry correction to a global similarity* — learned from each point’s own neighborhood. It is the cheap static cousin of the per-entry threshold: the field discovered twice, a decade apart, that one number cannot govern every neighborhood.

²⁵ Zhang, Suya, Jha, Zhang, Shmatikov, *Adversarial Hubness in Multi-Modal Retrieval*, IEEE S&P 2026 (arXiv:2412.14113).

²⁶ The two-sided counting result is the Kabatjanskii–Levenshtein bound; Tao’s 2013 exposition (“a cheap version of the K–L bound”) gives the $e^{c\varepsilon^2d}$ statement cleanly; Vershynin, HDP 2018, §3.3 covers the packing machinery; Cai, Fan, Jiang, JMLR 2013 give the random-vector angle distributions.

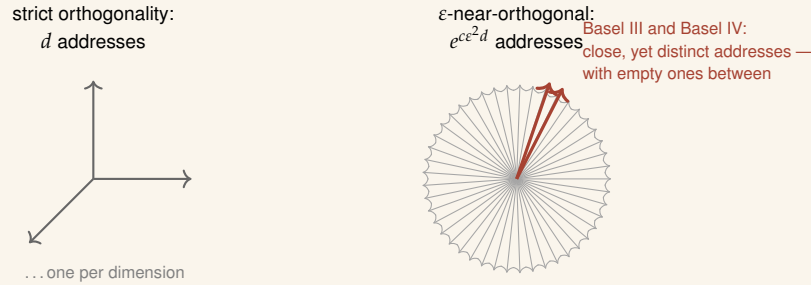


Figure 8: The sphere’s address book. Demand exact orthogonality and you get d addresses; tolerate a whisker of cosine and the count explodes exponentially. Near-identical meanings are never forced to collide — every collision is an assignment the encoder made, and fine-tuning can revoke.

²⁷ Elhage et al., *Toy Models of Superposition*, Transformer Circuits / Anthropic, 2022 (arXiv:2209.10652) — explicitly the near-orthogonal packing argument. The taught figure: fig-superposition-unification, RAG course. Recent formal capacity treatments: arXiv:2409.15318 (2024); arXiv:2602.11246, arXiv:2606.02765 (2026).

The capacity theorem is also the quiet guarantor of the metropolis picture: downtowns can be arbitrarily dense *and* internally distinguishable, because even within a tight cap there are exponentially many sub-addresses. Density limits the *encoder’s resolution*, never the sphere’s.

This is the same mathematics that interpretability rediscovered as **superposition** — networks storing far more features than dimensions by assigning them nearly-orthogonal directions — and the Session-9 corpus already owns the figure for it.²⁷ For the cache, the capacity theorem settles a question of blame with the finality of an audit: *collisions are never a capacity problem*. When Basel III and Basel IV sit at cosine 0.96, the sphere did not run out of room — exponentially many addresses stood empty between them. Their proximity is a *choice* made by the encoder’s training data, and it is revisable: the whole fine-tuning pipeline of the implementation chapter (the Basel similarity dropping from 0.96 to 0.72 after domain tuning) is, in this light, the act of moving two tenants who were wrongly assigned one address into the separate quarters the sphere always had for them.

The Aperture Doctrine, Finished

STEP BACK AND ASSEMBLE THE FINISHED PICTURE, because the four stories are one story. The previous chapter’s theorem built the wall against chance: nothing enters the cap accidentally. This chapter surveyed the city inside the walls. The manifold is small inside its container (intrinsic dimension), so contrast survives and nearest-neighbor means something. Fine-tuning does not flatten the city; it *builds* it — metropolises of meaning spaced by uniformity, packed by alignment, zoned by temperature. The city has an aristocracy (hubs) that demands per-neighborhood correction, and now an adversary who campaigns to join it. And the city can never run out of addresses — only out of encoder resolution to use them. Every one of the decision chapter’s conclusions re-derives from this geography: per-entry thresholds because zoning is local; calibration because downtowns differ; verification because aristocrats lie; scoping because impostors campaign. The threshold was promoted from a number to a policy in the last chapter’s language of statistics;

this chapter says the same thing in the language of the sphere: *a city is not governed by a single rule, and the cache is the city's gatekeeper, not its geometer.*

The Decision: When the Cap Is Not Enough

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PART II ENDED WITH AN UNPRICED PROBLEM: judgment among the bears. The geometry guarantees that everything above threshold was placed there by learning; it declines to say whether what was placed there means the *same thing* as your query. For three cohorts we have answered that question with a single number — sweep a labelled set offline, draw the precision–recall curve, pick τ , ship it. The sweep is honest work and remains the right first lecture. But it makes a silent assumption so natural that we never said it aloud: *that one threshold can be right for every entry in the cache*. The founding measurement of the modern literature is that this assumption is false — not slightly false at the margins, but structurally false, entry by entry — and everything in this part flows from taking its falseness seriously.

The Overlap, Measured

HERE IS THE EXPERIMENT OUR SWEEP NEVER RAN. Take a deployed cache. For every *hit* it serves, record two things: the similarity score that authorized the hit, and — by paying for the LLM call we were trying to avoid, on a sampled subset — whether the served answer was actually correct for the new query. Now plot the similarity distributions of the

²⁸ Schroeder et al., *vCache: Verified Semantic Prompt Caching*, arXiv:2502.03771, ICLR 2026. The per-embedding overlap of correct- and incorrect-hit similarity distributions is the paper’s founding observation; everything else in it is machinery for living with that fact.

correct hits and the incorrect ones. If the single-threshold picture were true, the two distributions would separate: wrong hits low, right hits high, and τ a clean wall between them. What is actually observed is *overlap* — heavy, irreducible overlap — and, worse for the sweep, overlap whose shape *differs per cached entry*. Around one cached question, similarity 0.91 is reliably safe; around another, 0.96 still serves wrong answers, because that entry sits in a crowded neighborhood where its hard negatives live at 0.97.²⁸

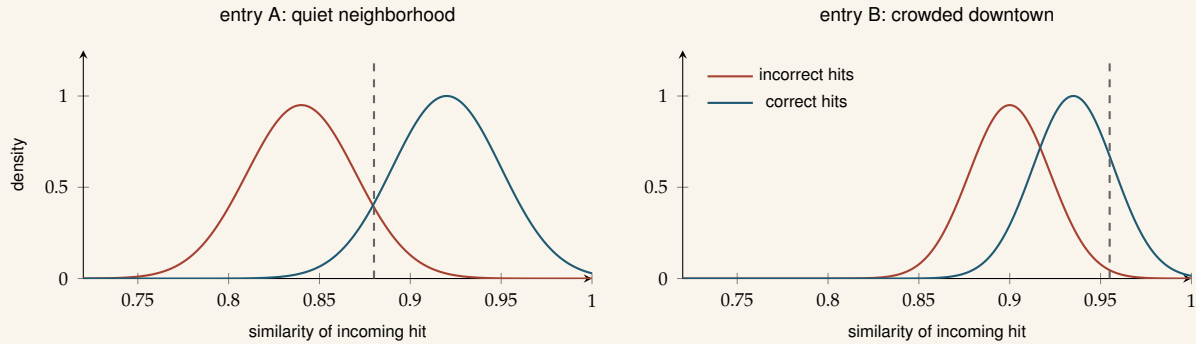


Figure 9: The founding measurement of the modern literature, schematically. Correct- and incorrect-hit similarity distributions overlap — and overlap differently per cached entry, because within-cap geometry is local. Entry A is safe at 0.88; entry B still serves wrong answers at 0.94. No global threshold serves both; the sweep merely averages the neighborhoods. Hence: per-entry decision curves under an error budget.

The taught sweep retains one virtue the new machinery inherits rather than replaces: it forces the team to build a labelled set of paraphrase/near-miss pairs. Every method in this part consumes such labels; the sweep at least made us collect them.

Part II tells us we should not have been surprised. Within-cap structure is governed by the local accidents of the encoder’s training, and “local” means *local*: the density of hard negatives around “Basel III capital requirements” has no reason to match the density around “how do I reset my password.” A global τ is a single answer to a question that is asked differently in every neighborhood of the sphere. The offline sweep does not fix this; it *averages* it — choosing the threshold that is least wrong on average across neighborhoods, which is to say, wrong nearly everywhere by a little, and badly wrong exactly where the workload is dense and the stakes concentrate.

The Serve Decision, Priced in the Family Currency

BEFORE THE MACHINERY, THE ECONOMICS OF A SINGLE HIT, stated in the equation family this corpus already owns — because the serve decision is the third sibling of two decisions we have priced before. SKILLCRAFT priced loading a skill: $\text{VoL} = p g - \kappa(c)$. MEMORYCRAFT priced recalling a memory: $\text{VoR}(m) = p_m g_m - \kappa(c_m)$, character for character the same law. The cache prices *serving* a stored answer, and the new term is the

one that makes semantic caching uniquely dangerous — the cost of being *wrong*, not merely unhelpful:

$$\text{VoS}(i \mid q) = p_i g - (1 - p_i) L - \kappa, \quad \text{serve } i \iff \text{VoS}(i \mid q) > 0,$$

where $p_i = \Pr[r_i \text{ correctly answers } q]$ is the probability this hit is right, g the value of a correct instant answer (saved output tokens, saved seconds), L the loss from serving a wrong answer with confidence, and κ the small serving overhead (embed, search, verify). Rearranged, the serve condition is a threshold on *probability*, not on similarity:

$$\text{serve} \iff p_i > \frac{L + \kappa}{g + L} \approx \frac{L}{g + L}.$$

Read it aloud: *how sure you must be is set by the ratio of harm to help* — a compliance workload with crushing L demands p_i near one; a low-stakes FAQ tolerates far less. Similarity appears nowhere in this law. Similarity is merely the *evidence* from which p_i must be estimated — and the overlap measurement says the map from evidence to probability differs per entry. The entire modern program falls out of this one rearrangement: *estimate p_i per entry, honestly, and let the error tolerance — not the cosine — draw the line.*

The Threshold Learns: Verified Caching

THE REFERENCE DESIGN FOR ESTIMATING p_i IS vCACHE, and its shape is worth teaching as three moves, because each move answers one of the sweep’s failures.

Move one: the contract inverts. The operator no longer chooses a similarity; the operator states an error budget: “of the hits this cache serves, at most a fraction δ may be wrong.” One in fifty; one in five hundred for the regulated floor — the number is a policy decision, the $L/(g + L)$ of the previous section wearing operational clothes. Everything else becomes the cache’s problem, with the budget as a contract it must be able to demonstrate it is keeping.

Move two: every entry keeps its own books. For each cached embedding, the system maintains a little model — a one-parameter sigmoid family suffices — of $\Pr[\text{correct} \mid s]$, the probability that a hit at similarity s against *this entry* is right. Crowded neighborhoods learn steep, late-rising curves; lonely entries learn generous ones. The global τ dissolves into a per-entry decision boundary, each placed exactly where that entry’s own evidence puts it.

Sibling-rhyme ledger: VoL (SKILLCRAFT) $\rightarrow$ VoR (MEMORYCRAFT) $\rightarrow$ VoS (here). One law, three doors. The new $(1 - p)L$ term is not a generalization for its own sake; it is Part I’s principle — *only the semantic cache can be wrong* — entering the algebra.

²⁹ vCache, *ibid.*: up to $12.5\times$ higher hit rate at $26\times$ lower error than static-threshold and fine-tuned-embedding baselines, across three embedding models and two LLMs, with the error bound δ honored throughout. Open-source; four public benchmarks released with the paper.

Move three: exploration pays for the labels. Where do correctness labels come from, online, without a human? From spending, judiciously, the very calls the cache exists to save: with a calibrated probability the system serves a would-be hit *and* silently invokes the model anyway, compares the two answers, and books the agreement or disagreement as a label for that entry's curve. Early in an entry's life it explores often and serves cautiously; as evidence accumulates, exploration decays and the entry earns its autonomy. The result, across benchmarks and embedders, is not a modest improvement but a change of regime: the stated error bound is actually met, at hit rates up to an order of magnitude above what any compliant static threshold could reach.²⁹

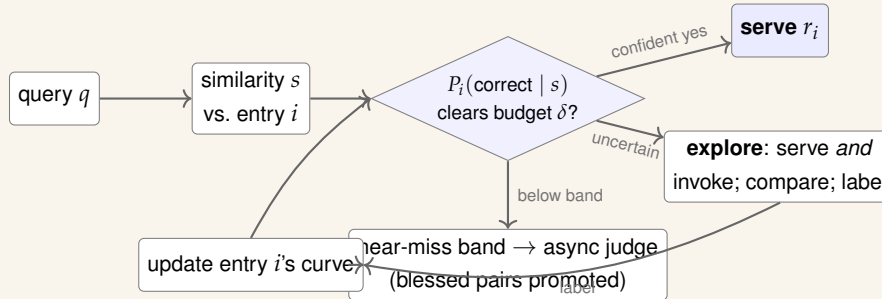


Figure 10: The verified serve path. Similarity is evidence, the per-entry curve is the judge, the budget δ is the law, exploration pays for the labels, and the near-miss band gets an asynchronous second opinion. Proposal plus gate, drawn.

Principle 0.3 (State the error you can afford; make the cache prove it). A similarity threshold is an implementation detail. The designed object is the error contract: a stated bound δ on the fraction of served hits that may be wrong, kept per entry by learned decision curves and paid for by measured exploration. If a cache cannot tell you its demonstrated error rate, it does not have a threshold — it has a guess.

NOTICE WHAT THIS ARCHITECTURE IS, STRUCTURALLY: it is the course's oldest invariant wearing cache clothes. A proposal (serve this hit) faces a gate (the per-entry curve under a fixed budget), and the gate is maintained by an eval that never stops running (exploration as perpetual, sampled ground truth). Proposal plus gate; self-improvement under a fixed judge — GEPA's law, the skill governor's law, the memory governor's law, now the cache's law. When you teach this, teach it as the fourth instance of the one architecture, because that is what it is.

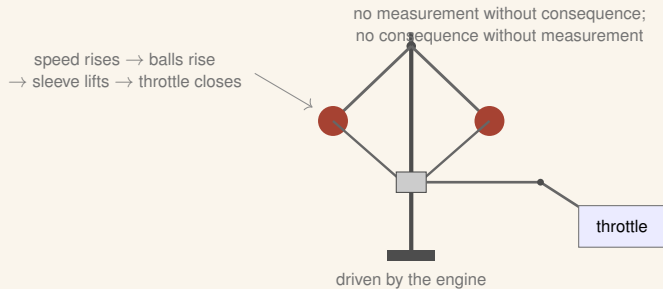


Figure 11: The ancestor of every gate in this corpus: Watt's centrifugal governor. The engine proposes a speed; the spinning flyballs judge it and physically close the throttle. Proposal plus gate, in brass — and the subject of Maxwell's *On Governors* (1868), the founding paper of control theory.

Calibration: Measuring the Right Thing

A QUIETER 2026 RESULT EXPLAINS WHY GOOD OFFLINE NUMBERS KEPT PRODUCING BAD CACHES, and it deserves a permanent place in the evals-week canon. The field's habitual offline metric for cache matchers is PR-AUC on a labelled pair set — exactly our taught sweep's summary statistic. PR-AUC rewards *ranking*: correct pairs above incorrect pairs, anywhere on the score axis. But a cache does not rank; it *thresholds*, and thresholding needs the scores themselves to be usable — spread out, stable, meaning the same thing across the range. A model can rank superbly while squeezing all its scores into an uninterpretable sliver, and the measured collapse is dramatic: rerankers trained with binary cross-entropy, PR-AUC 0.75–0.82, retain less than a quarter of that performance when actually thresholded — while ColBERT-style late-interaction scorers, mediocre rankers on paper, threshold almost perfectly. Most striking of all: thirty-eight times more training data *worsened* deployment behavior when the objective was wrong. Calibration is set by the loss function, not by scale.³⁰

Three consequences for this house. *First*, a small vindication: the taught fine-tuning pipeline optimizes InfoNCE — a contrastive objective — which lands on the calibration-friendly side of the divide; our recipe was better than our metric. *Second*, a correction: the sweep's report card must change. Report the *precision-versus-hit-ratio curve* — for each operating point, what fraction of traffic is served and what fraction of served answers is right — and its area, not PR-AUC; it is the curve an operator actually lives on. *Third*, a retirement: the corpus's home-made freshness metric MRF, defined one way in the chapter and another in the lesson plan, is hereby superseded rather than repaired. The field now has sharper instruments for everything MRF gestured at: the

³⁰ Baral, Ralev, Zhechev, Rajamohan, Agarwal, *Closing the Calibration Gap in Semantic Caching*, arXiv:2606.19719 (2026): the P-CHR curve (precision versus cache-hit ratio), its AUC, and the Calibration Retention Rate = P-CHR AUC / PR-AUC; BCE-trained rerankers collapse to 0.17–0.20 P-CHR; contrastive (MNRL) objectives fare better; late-interaction scores calibrate best. Redis-affiliated and methodological — it complicates the vendor story rather than flattering it.

Erratum candidate #4, resolved by retirement. The four-metric dashboard survives with one seat re-assigned: hit rate, FPR (now: demonstrated error against δ), unsafe-served rate, and latency-savings ratio.

³¹ Krites, arXiv:2602.13165 (2026): asynchronous judge-verification of the near-miss band, promoting verified pairs into a dynamic tier; up to $3.9\times$ more requests served from curated answers at unchanged critical-path latency.

³² MVR-cache, arXiv:2605.24914, ICML 2026: learned prompt segmentation + late-interaction matching under the vCache guarantee; up to +37% hit rate over vCache; needs $\sim 3k$ labelled prompts per domain; ~ 23 ms segmentation overhead.

unsafe-served rate for freshness failures (Part V, with the version-aware gates that drive it to zero) and the P-CHR curve for threshold quality. A metric that needs a reconciliation footnote has told you its replacement is due.

The Verification Family

LEARNED THRESHOLDS ARE ONE ROAD TO HONEST HITS; THE OTHER IS TO STOP TRUSTING SIMILARITY ENTIRELY AND *check*. The verification family spends a little compute per hit to convert evidence into assurance, and its members differ only in where the check runs. *Asynchronous verification* (Krites) keeps the fast path untouched: hits above the confident line serve instantly, while the near-miss band — just below threshold, where the overlap lives — is shipped, off the critical path, to a cheap LLM judge; pairs the judge blesses are promoted, so the cache *curates itself* without ever charging the user latency.³¹ *Synchronous verification* pays the latency where wrongness is unaffordable: before serving, a fast check that the cached answer actually addresses the new query — the voice-AI shops run exactly this, having learned from pairs of credit-card questions at cosine 0.99 with different correct answers that in some domains *no* similarity is high enough. *Structural verification* changes the evidence itself: multi-vector matching (MVR-cache) segments the prompt with a learned segmenter and matches segment-by-segment, MaxSim-style, so that negation, direction, and slot values — the classic single-vector blurs — become visible before any threshold is consulted; it slots into the vCache framework and provably preserves the error guarantee.³²

The modern serve path, assembled

Evidence: similarity — per-segment where the domain blurs (MVR). Decision: per-entry curve under a stated budget δ (vCache), with δ chosen by $L/(g + L)$. Assurance: asynchronous judging of the near-miss band (Krites); synchronous checking where L is catastrophic. Bookkeeping: P-CHR reported, exploration labels accumulating, error contract demonstrated weekly. Every component exists, is published, and is open-source; none of it was available when the Session-9 chapter was written. This box is that chapter's upgrade path.

The Ladder, in Practice

THE CLASSROOM NEEDS A LADDER, NOT A LEAP, because the taught sweep is not wrong — it is rung one. *Rung one*: global τ from a calibrated sweep, P-CHR reported, weekly graded samples — fully appropriate for low-stakes, high-repetition FAQ traffic, and the only rung most gateways even offer. *Rung two*: per-category thresholds and TTLs — code queries, chatty queries, regulated queries each get their own operating point; cheap, shippable today, and already the production norm at scale. *Rung three*: the error contract — per-entry learned curves with exploration, for workloads dense enough to feed them. *Rung four*: verification, async first, sync where L demands. Each rung strictly dominates the one below when its preconditions hold; the preconditions are label supply and traffic density; and the honest instruction to the class is the one this part has been building to: *choose your rung by your $L/(g+L)$, and whatever rung you stand on, be able to state your demonstrated error rate — because a threshold you cannot audit is a guess you have automated.*

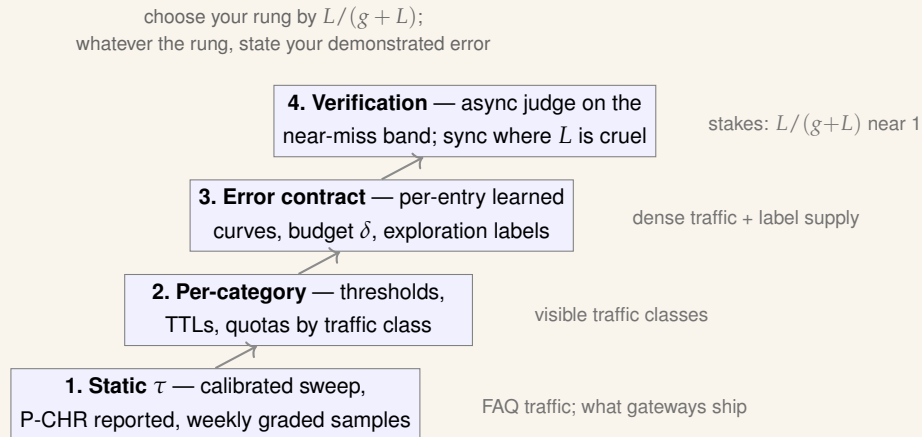


Figure 12: The threshold ladder. Each rung strictly dominates the one below when its precondition (right) holds. Shipped products live on rung one; the 2025–26 literature lives on rungs three and four; the gap between them is the most exploitable fact in this report.

Step back once more. The geometry of Part II built a wall against chance; this part built a court for judgment: a price on being wrong, a per-entry estimate of being right, a budget the cache must prove it keeps, and checkers for the band where evidence runs thin. The two sacred objects survive enlarged — E now trained for calibration, not just separation; τ now a policy with a contract, not a number with a hope. What remains is machinery: the encoder itself (Part IV), the question of what exactly to store (Part V), and the agent workloads where similarity was never the right evidence at all (Part VI).

Implementation: The Modern Lookup Stack

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PARTS II AND III SPENT THEIR PAGES ON τ ; THIS PART BELONGS TO *E*. The taught corpus is at its strongest here — the Session-9 fine-tuning pipeline, from domain-blindness diagnosis through InfoNCE with mined hard negatives to distillation and threshold recalibration, is real engineering craft and survives this report intact. What has changed since it was written is not the pipeline’s shape but three of its assumptions: *which* encoder to start from, *what objective* to train it toward, and *what the key should even be* when queries arrive trailing conversations behind them. This part updates each assumption against the 2026 evidence, then descends into the machinery beneath the encoder — the index, the metadata, the warming, and the eviction — where several of our margin-level treatments now have a real literature to stand on.

The Encoder, Purpose-Built

BEGIN WITH A VINDICATION, BECAUSE IT IS EARNED. The taught pipeline fine-tunes ModernBERT-class encoders on domain pairs, and the question that commissioned this part — *is that still best practice?* — has a clean answer from an unexpected witness. Redis, building its managed cache product, spent 2025 training purpose-built cache embedders, and

³³ Redis langcache-embed line: v1 (April 2025, with the struggled-to-beat admission in the accompanying write-up); v2; v3-small (January 2026) — ~20M parameters, 128-token inputs, trained on 8M+ labelled question pairs specifically to separate “looks-different/means-same” from “looks-similar/means-different.” Vendor-published; no hard benchmark numbers in the v3 post.

³⁴ Gill et al. (Redis + Virginia Tech), *Advancing Semantic Caching for LLMs with Domain-Specific Embeddings and Synthetic Data*, arXiv:2504.02268 (2025). The synthetic-pair pipeline is the paper’s practical gift: a generator for exactly the labelled paraphrase/hard-negative pairs the taught pipeline requires 5,000 of.

Part III, §. The sharpest cautionary number bears repeating at the keyboard: in the calibration study, *thirty-eight times more training data made deployment behavior worse* under the wrong objective. Scale is not a substitute for the right loss.

³⁵ The source treatment: *Encoder Fine-Tuning*, fine-tuning course — the hard-negative-mining section, the curriculum-learning doctrine, and the orthogonality-of-method-and-loss section. This subsection restates only what the cache changes; the derivations live there.

their own published account of the first attempt records that it *struggled to outperform* an off-the-shelf gte-modernbert-base.³³ The academic companion result says why the ModernBERT-plus-fine-tuning recipe holds: small embedding models fine-tuned for even one epoch on domain data — real pairs supplemented by synthetic paraphrase/near-miss generation — beat much larger general-purpose and proprietary embedders on cache-matching precision and recall, at a fraction of the latency.³⁴ The cache-matching problem is narrow — *is this question the same question?* — and narrow problems reward small specialists over large generalists. The instinct, taught for three cohorts, ran ahead of the vendors who eventually confirmed it.

Two genuine updates, then. *First, the objective is now chosen for calibration, not only separation.* Part III’s result bears directly on the training loop: a matcher trained with binary cross-entropy can rank beautifully and threshold uselessly, while contrastive objectives — InfoNCE among them — produce score distributions an operator can actually cut. The taught pipeline was on the right side of this divide by instinct; it should now be on the right side *on purpose*: evaluate every fine-tune on the precision-versus-hit-ratio curve, and if a reranking stage is added atop the bi-encoder, prefer late-interaction scoring, whose calibration retention is the best measured. *Second, the starting checkpoint can now be a cache specialist.* Where domain data is scarce — a young product, a cold start — the purpose-built cache embedders (20M parameters, question-pair-trained) are a legitimate starting point that did not exist when the chapter was written; fine-tune *from* them rather than from a general checkpoint, and the 5,000-pair entry fee drops accordingly.

The Curriculum, the Coupled Knobs, and the Orthogonal Axes

THE TRAINING SCHEDULE ITSELF IS DOCTRINE IN THIS HOUSE, taught in full in the fine-tuning course’s encoder chapter, and the cache inherits it with three report-specific sharpenings.³⁵ The curriculum first: not all negatives are created equal, and their difficulty must be *staged*. Round one trains against the full batch — easy to moderate negatives, the bulk separations. Round two mines: embed the corpus with the partially trained model, harvest for each anchor the *k* nearest non-positives — the pairs the model *struggled to repel* — and train against those alone. Iterate if the eval demands it; diminishing returns arrive quickly. For the cache, the curriculum’s stages map cleanly onto the label supply chain of Part III: synthetic paraphrase pairs are the easy round, log-mined pairs the moderate one, and the *near-miss band* — the cache’s own

just-below- τ diagnostic stream — is the hard round, arriving pre-mined by production itself. And each stage is now graded on the deployment curve (P-CHR), not the ranking curve, because the calibration result says the curriculum’s later, harder stages are exactly where a wrong objective does its silent damage.

THE TAUGHT CHAPTER ALSO NAMES THE TEMPTATION, AND THE CACHE PAYS ITS PRICE TWICE OVER. Mine too aggressively and two pathologies follow: the positive pairs collapse together, crushing nuance within the class; and the *somewhat-related* — items a user would have wanted — get expelled beyond the polar cap entirely. In retrieval that empties search results; in the cache it is subtler and worse: the over-mined encoder silently narrows every cap, the hit rate sags for no visible reason, and the partial-match tier — which lives precisely on the somewhat-related — starves. The Basel III/IV pair is the target discrimination; the queries about Basel *generally* are the collateral, and an encoder that can no longer see their kinship has traded the cache’s revenue for its precision. The temperature–threshold coupling deserves promotion from margin to principle, because it closes a loop this report has been circling since Part II: *every fine-tune redraws the map that τ reads*. Retrain the encoder — new temperature, new mining round, new checkpoint — and every threshold calibrated against the old geometry is silently wrong: the folklore 0.85 means a different angular radius on the new sphere. The operational consequence is the Version Pin pattern wearing its training-time face: an encoder change is a threshold-recalibration event, unconditionally — re-sweep on rung one, re-learn the per-entry curves on rung three — and a fine-tune shipped without recalibration is an invisible threshold change shipped to production.

The fine-tuning chapter’s deepest margin note belongs in the cache canon verbatim: *temperature and threshold are intimately connected*. A low training temperature produces tight clusters — and narrows the effective polar cap that any given τ selects; a high temperature widens it. The two knobs must be reasoned *jointly*, and the geometric question — how wide should my cap be? — is the tractable form of both.

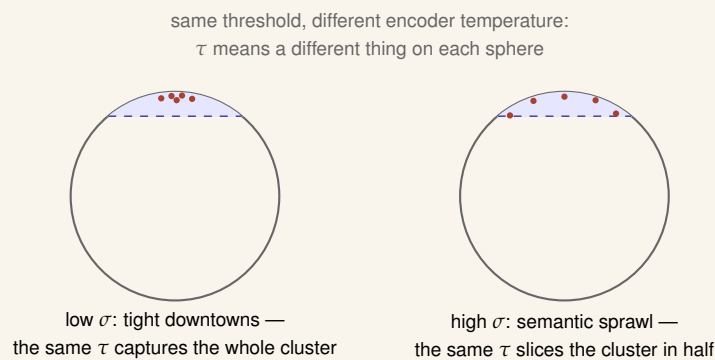


Figure 13: The temperature–threshold coupling, drawn. Training temperature zones the downtowns; the threshold selects a cap. Retune one without recalibrating the other and you have shipped an invisible threshold change. Every fine-tune is a recalibration event.

LAST, THE ORTHOGONALITY DOCTRINE — THE DECOUPLING THAT KEEPS THE DESIGN SPACE NAVIGABLE. The choice of contrastive objective (triplet,

InfoNCE, SimCLR-style) and the choice of adaptation strategy (full fine-tune, LoRA, last-layer) are *independent axes*: what forces to apply, and how many parameters may move. Every cell of that matrix is a legitimate configuration, chosen by data volume, domain shift, and budget. For the cache-specialist encoder of this part, the typical cell is small-model $\times$ (LoRA or last-layer) $\times$ InfoNCE-with- curriculum — the smallest configuration that clears the P-CHR bar — and the decoupling is what lets a team upgrade its loss without re-litigating its compute, or vice versa, one axis at a time.

The Key Grows a Memory: Multi-Turn and Context

THE TAUGHT CACHE KEYS ON A CANONISED SINGLE QUERY, AND THE COUPLING DOCTRINE — CACHE AFTER TRANSFORMATION, ALWAYS — SURVIVES UNTOUCHED. What the 2025–26 literature adds is the case our chapter set aside: the query that cannot be canonised without its conversation. “Does that still work for me?” embeds into six pronoun-laden tokens; identical surface queries in different dialogues demand different answers; and a cache blind to history will serve one user’s context to another user’s words. Two architectural answers now exist. *Context-integrated matching*: retrieve candidates by the current query, then score the match with an attention mechanism over the *dialogue representations* of both sides, so that surface-identical queries in different conversations separate — at latency still an order of magnitude below generation.³⁶ *Context-partitioned keys*: fold the stable frame — system prompt, tenant, persona, tool schema — into an exact hash, and embed only the volatile final message; conversation chains link per-user entries so follow-ups resolve against their own thread. This is the production pattern (it is also, not coincidentally, how the per-user private caches of the federated line work), and it composes with the taught access-partition doctrine: the cabinet gets one more drawer key, the conversation.³⁷

The taught *cache-eligibility classifier* — temporal, personal, exploratory, and high-stakes queries bypass the cache — also acquires its production-scale form: category-aware caching, where each traffic category carries its own threshold, TTL, and quota, because code queries cluster dense and stale slowly while conversational queries scatter and rot fast. One knob per workload, not one knob per cache.³⁸

³⁶ Yan et al., *ContextCache*, arXiv:2506.22791, PVLDB 2025: two-stage retrieval + self-attention integration over current and historical turns; +10.9% precision, +14.8% recall over context-blind matching.

³⁷ MeanCache (arXiv:2403.02694) for per-user caches with context chains; the hash-the-frame, embed-the-message pattern is the one credible gateways document.

³⁸ Category-aware semantic caching, arXiv:2510.26835 (2025): per-category thresholds/TTLs/quotas; hybrid in-memory HNSW + external store dropping miss-path cost from ~ 30 ms to ~ 2 ms. Single-deployment evidence; the design pattern, not the numbers, is the takeaway.

The Machinery Below: Index, Metadata, Warmth

THREE PRACTICALITIES OUR TEACHING CARRIED IN MARGINS NOW GET THEIR PARAGRAPHS. *The index.* The ANN layer contributes the second, smaller term of the false-hit budget, and the latency arithmetic the gap analysis worked out deserves promotion into the canon: at cache scale the *embedding call dominates* end-to-end lookup — single-digit milliseconds against sub-millisecond HNSW search — which has two consequences: the small-specialist encoder of § is a latency decision as much as a quality one, and an exact-match tier in front (the taught Tier 1) is not an optimization but a necessity, since it spares the embedding call entirely on the head of the distribution. *The metadata split.* The week-9 errata’s clarification is hereby doctrine: the vector index holds vectors and IDs; *everything else* — responses, provenance, versions, decay clocks, access groups — lives in an auxiliary key-value store keyed by ID. Duplicating payloads into the index couples your invalidation to your index rebuild, which is exactly the coupling the version-pin rule exists to prevent. *The warmth.* The taught warm-start protocol — replay a month of logs, verify the embedder on the replayed neighbors — gains a predictive sibling: pre-populating the cache with *anticipated* queries, enumerated by a model under a likelihood bound, which converts part of the semantic matching problem into exact matching over a predicted question space.³⁹ Between replay and prediction, a new cache need never open cold.

³⁹ InstCache, Zou et al., arXiv:2411.13820: NLL-bounded instruction enumeration for predictive pre-population; 2.3× hit rate over reactive caching on open-chat workloads.

Eviction, Revisited by Theory

OUR ONE-PARAGRAPH TREATMENT OF EVICTION — LRU, LFU, ARC, CHOOSE SENSIBLY — HAS BEEN OVERTAKEN BY A SMALL, SHARP LITERATURE, and its headline results rearrange the furniture. First: *Belady does not survive approximate hits.* The offline-optimal eviction that anchors every classical cache argument (evict the entry whose next use lies farthest in the future) is well-defined only when a request hits exactly one entry; under similarity matching, one arrival can be served by many entries and one entry serves many arrivals, the problem becomes a coverage problem, and computing the offline optimum is NP-hard. The clean classical north star our MEMORYCRAFT cache canon leaned on simply does not exist here.⁴⁰ Second, and more startling: *the classical online policies lose to FIFO* on semantic-retrieval workloads — LRU’s temporal-locality assumption and LFU’s stationary frequencies both

⁴⁰ Biton, Friedman (Technion), *From Exact Hits to Close Enough*, arXiv:2603.03301 (2026): offline-optimal replacement under similarity hits reduces from Maximum Coverage; heuristics plus SphereLFU online; across nine datasets, frequency signals beat recency signals.

⁴¹ SOLAR, Sun, Cao, Lam, arXiv:2607.00394, PVLDB 2026: LRU/LFU/ARC underperform FIFO on semantic buffers; learning-augmented replacement with competitive ratio ≤ 3 independent of cache size.

Sibling-rhyme note for the ledger: MEMORYCRAFT’s cache canon (Bélady, LRU/LFU/ARC, ghost lists) stands for context eviction, where a memory either is or is not re-attended. CacheCraft adds the corollary, not a contradiction: *approximate* hits break the canon’s ground rules, and the replacement theory must be rebuilt — a rebuild now underway in the literature.

fail when hits are continuous-valued and neighborhood-shaped, and a learning-augmented policy (regret-gated admission, sampled utilities with temporal decay) is the first to beat naive baselines with a guarantee.⁴¹ The practical residue for a working cache is modest and worth stating plainly: prefer frequency-flavored scores over recency-flavored ones; treat admission (should this write enter at all?) as seriously as eviction — the taught salience gate was always half of replacement policy; and hold the classical vocabulary a little more loosely here than MEMORYCRAFT could afford to, because this is the one cache in the family where the old theorems’ ground rules genuinely changed.

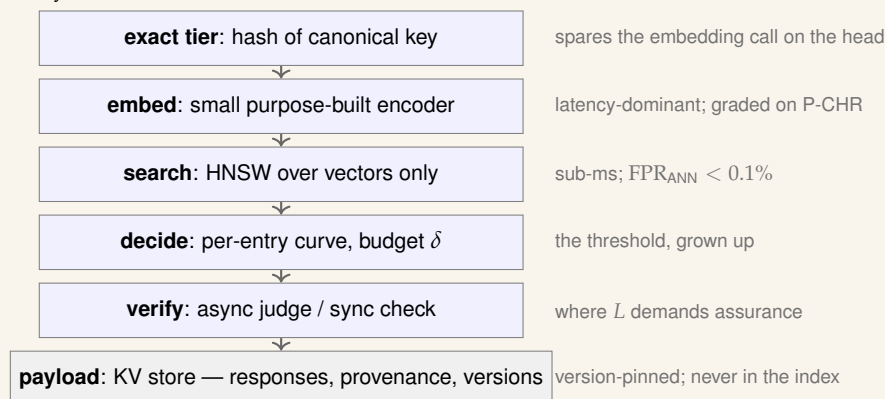


Figure 14: The 2026 lookup stack, assembled. Every layer existed in some form in the taught material; the field’s contribution since is the evidence for which layer matters and the theory for why.

STEP BACK. The lookup stack of 2026, assembled: an exact-match tier on the head; a small, purpose-built, contrastively trained, calibration-graded encoder; context folded into the key by hash or by attention; an HNSW index holding nothing but vectors; a key-value store holding everything else, version-pinned; warmth by replay and prediction; admission gated by salience; eviction scored by frequency and utility, under no illusions about optimality. Every box in that stack existed in some form in the taught material; what the field supplied since is the evidence for which box matters and the theory for why. The next part asks a question the stack takes as given — *what, exactly, should the entries be?* — and the answer turns out to be five-fold.

What to Cache in a RAG System

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OUR TEACHING USED “THE CACHE” IN THE SINGULAR, and the definite article was doing quiet, unexamined work. The Session-9 placement figure draws one valve on the pipeline: after query transformation, before retrieval, holding finished answers. That valve is real and remains the front door. But walk the RAG pipeline slowly — query, retrieval, assembly, generation — and count the joints where an expensive artifact gets recomputed for a repeating input: there are five, and the field has now built a distinct caching literature at each. The question for a working system is no longer *whether* to cache but *which layer to cache at* — and the layers differ enormously in what they save, what they risk, and how they die.

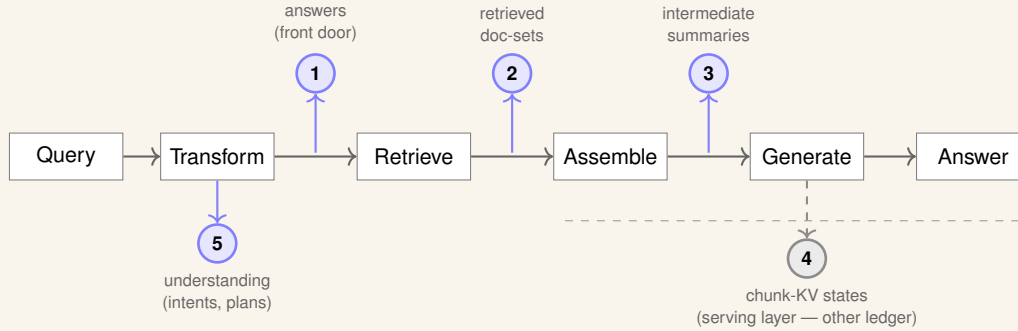


Figure 15: What a RAG system can cache: the five valves. Valve 1 is the taught front door — the only valve that can serve a wrong answer. Valves 2 and 3 reuse evidence and middles while generation stays fresh. Valve 5 caches meaning itself. Valve 4 lives below the dashed line, at the serving altitude, and posts to a different ledger entirely.

Layer One: Answers — the Front Door

THE TAUGHT CACHE: QUERY IN, FINISHED ANSWER OUT, bypassing retrieval *and* generation — maximum savings, maximum danger, and by every honest measurement the *lowest* hit rates of any workload class, because RAG queries are long-tail by construction: the whole reason an organization builds retrieval is that its people ask questions no FAQ anticipated. Its distinctive pathology also lives here: the answer cache *freezes generations against a moving index*. Re-chunk the corpus, update a policy document, swap the generator model — and every cached answer derived from the old world is now a confident fossil. The taught invalidation triple (TTL, document-change triggers, confidence decay) plus the version pin were built for exactly this layer, and § upgrades them with the machinery the field added.

Layer Two: Retrieved Sets — the Safer Bet

ONE VALVE UPSTREAM SITS THE CACHE ALMOST NOBODY SHIPS AND MORE TEAMS SHOULD: key on the query, store the *retrieved document set* — IDs and scores, not prose. A hit skips embedding, search, and reranking, then *generates fresh* over the cached evidence. The savings are smaller (retrieval latency, vector-database load) but the correctness risk collapses: generation never staled, and invalidation becomes exact — a document version changes, every cached set containing its ID dies, by index lookup rather than semantic guesswork.⁴² On skewed workloads this layer removes three-quarters of retrieval traffic while keeping every answer current — and because it stores IDs, its entries are tiny. That no

The practitioner ladder from Part I, restated for this layer alone: RAG-backed Q&A observes 15–25% hit rates in production — against 40–60% for FAQ-style support traffic. The front door earns its keep on the head of the distribution and must be *allowed to miss* on the tail.

The revision of mid-August 2026 adds a fourth invalidator at the chapter's end: the *authored* expiry — an absolute date stamped at write time by the author who knows the fact's shelf life, rather than a relative countdown guessed by the operator who caches it (§).

⁴² Bergman et al., *Proximity*, arXiv:2503.05530, Middleware 2025: approximate caching of retrieved sets with LSH-fast lookup; 77.2% fewer vector-database queries on skewed workloads at maintained answer accuracy.

major framework ships this as a first-class primitive is one of the small scandals of the 2026 stack; the teams that run it hand-roll it.

Layer Three: Intermediate Products

BETWEEN EVIDENCE AND ANSWER, PIPELINES INCREASINGLY MANUFACTURE MIDDLES — per-document contextual summaries, extracted tables, sub-question decompositions — and the middles are cacheable on their own terms: reusable across *similar* queries even when final answers must differ, because a good summary of a policy document serves every question that touches the document. Caching contextual summaries alone removes half to sixty percent of redundant computation on standard QA benchmarks at comparable answer quality.⁴³ The design insight generalizes: *the coarser the artifact, the wider its reuse and the safer its staleness profile*. A summary staled by an edit degrades an answer; a stale final answer *is* the degradation.

Layer Four: The Computation Itself — a Border Crossing

THE FOURTH LAYER THIS REPORT CAN ONLY WAVE AT, BECAUSE PART I FENCED IT DELIBERATELY: caching the *KV states* of retrieved chunks so the generator need not re-prefill them — per-chunk precomputation with selective recomputation to repair cross-attention, position-independent variants, offline cross-attention preprocessing.⁴⁴ It reuses computation, not answers; it cannot serve a wrong answer, only a slightly bruised one; it lives in the serving stack, not the application. It appears in this taxonomy for one reason: so that the placement map is complete and the boundary stays bright. MEMORYCRAFT taught two altitudes of eviction and commanded that the ledgers never mix; this is the same commandment, third appearance, and the CacheBlend erratum of Part I is what happens when it is broken even in a reading list.

The layer-two cache is the one to teach risk-averse shops first: it is the taught architecture with the danger amputated. RETRIEVALCRAFT's two-column ledger applies cleanly — it posts to the cost column and *cannot* post to the quality column, because generation always runs.

⁴³ Ramírez et al., *Semantic Caching of Contextual Summaries*, arXiv:2505.11271 (2025): 50–60% compute reduction on NaturalQuestions/TriviaQA-class workloads.

⁴⁴ The family: RAGCache (arXiv:2404.12457); CacheBlend, EuroSys 2025 (arXiv:2405.16444); TurboRAG (EMNLP 2025); EPIC (arXiv:2410.15332); FusionRAG, SIGMOD 2026 (arXiv:2601.12904), which also documents CacheBlend's token-selection bias — the literature auditing itself.

Layer Five: Understanding — Cache the Meaning, Not the Words

THE FIFTH LAYER KEYS ON THE QUERY AND STORES NEITHER ANSWER NOR EVIDENCE BUT *interpretation*: the resolved intent, the canonical structured form, the plan. The strongest production result in our corpus was always of this kind — the Instacart engine caching query *understanding* (the parsed intent for the ninety-eight percent head, precomputed offline) while answers stay live — and RETRIEVALCRAFT flagged it as the pattern to watch. The 2026 literature confirms the flag: for analytics, canonicalizing natural language into intent signatures gives *decidable* equivalence and zero spurious matches — the 1996 theorem rebuilt inside the LLM era; for agents, caching plans rather than responses is the only version that survives environment change. Both are Part VI's subject; the taxonomy notes here that they are not exotica but the fifth valve, and philosophically the deepest: it caches the one artifact — meaning — that repetition actually guarantees.

The five layers, on one card

Answers: biggest save, only layer that can be wrong, lowest RAG hit rates, hardest invalidation. *Retrieved sets*: medium save, cannot be wrong, exact invalidation — the risk-averse default. *Intermediates*: wide reuse across similar queries, graceful staleness. *KV states*: serving-layer computation reuse — different altitude, different ledger, never conflate. *Understanding*: decidable where schemas exist, the agent-safe unit, the deepest bet on repetition. A mature system typically runs two or three layers at once — and the taught single-valve figure was layer one's portrait, not the family's.

Freshness, Done Right

THE TAUGHT INVALIDATION TRIPLE WAS HONEST ENGINEERING WITHOUT A THEORY OF CORRECTNESS: TTLs guess at volatility, document-triggers over-kill (any edit murders every dependent entry, needed or not), and confidence decay lowers trust without checking anything. The field's best 2026 answer supplies what was missing: a *serial gauntlet*

of *cheap checks* that a cached answer must survive, at hit time, before it may be served. Query similarity first — the taught gate. Then *evidence overlap*: re-run cheap retrieval and require the fresh evidence set to overlap the cached provenance (the taught schema’s passage IDs, finally load-bearing). Then *version identity*: content-addressed chunk hashes — serve only if the evidence is byte-identical where it matters. Then, subtlest and empirically decisive, *lexical support*: verify the cached answer’s key claims still appear supported in the fresh evidence — catching the edit that kept the document’s identity but changed its meaning. On multi-hop benchmarks this gauntlet drives the *unsafe-served rate* — the fraction of hits that serve answers no longer supported by current evidence, the metric our retired MRF was groping toward — from double digits to zero, at four to seven percent added latency.⁴⁵

⁴⁵ Shah, *GroundedCache*, arXiv:2605.27494 (2026): four serial gates; unsafe-served 15–35% → 0.0% (HotpotQA), 26–52% → 1–10% (mtRAG), at 1.04–1.07× no-cache latency. The ablations show the lexical-support gate carrying the surprising share of the safety.

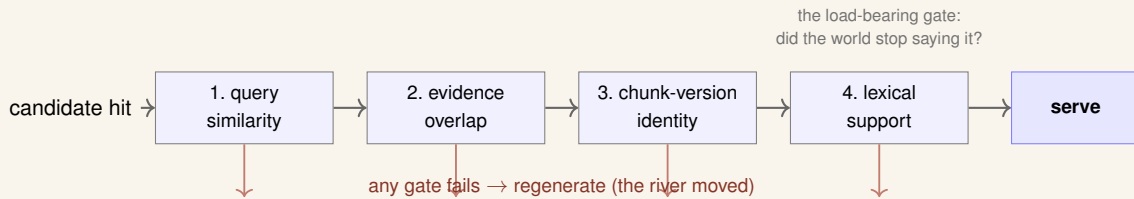


Figure 16: Freshness as a serial gauntlet, not a clock: similarity, then evidence overlap against cached provenance, then content-addressed version identity, then lexical support of the answer’s claims in fresh evidence. TTLs are demoted to backstops; the unsafe-served rate goes to zero.

Principle 0.4 (Invalidate by evidence, not by clock). A TTL encodes a guess about the world; provenance encodes a fact about the answer. Wherever the cached entry carries its evidence — and the taught schema always did — freshness should be *checked against that evidence at serve time*, with clocks demoted to backstops. The question is never “how old is this answer?” but “does the world still say what this answer assumed?”

ONE GENUINE FRONTIER REMAINS OPEN, AND THE LECTURE SHOULD SAY SO WITH RELISH: *negative caching*. Every treatment above caches successes; nobody has a principled treatment of caching the *absences* — the query that retrieved nothing, the question the system rightly declined, the miss itself (so the cache stops re-probing its own known gaps). Classical systems cache negative lookups as a matter of course; the semantic version — where a cached “we don’t know” must somehow expire when the corpus *learns* the answer — has no literature worth citing as of this writing. A capstone-sized hole, conveniently shaped.

STEP BACK. “What should the entries be?” turned out to be the design question the singular article was hiding: five artifacts, five risk profiles,

one bright border, and a freshness discipline that finally checks the world instead of the clock. The last workload standing is the one where even layer one's premise fails — the agent, whose repeated *words* conceal unrepeatable *states* — and it gets the next part to itself.

The Governed Layer: The Cache Above the Cache

This section entered the report in a revision of mid-August 2026, after OKFcRAFT — the ninth craft report — joined the shelf.

Nothing in it revises this chapter's doctrine; it seats, above the five layers, the one tier the placement map could not have drawn when it was inked, because the format that names it was eight weeks old.

⁴⁶ Sam McVeety and Amir Hormati, "How the Open Knowledge Format can improve data sharing," Google Cloud Blog, June 12, 2026.

Spec v0.2 and reference implementation: github.com/GoogleCloudPlatform/knowledge-catalog

The close reading this section leans on is OKFcRAFT, Parts II–III, whose vocabulary it borrows without re-deriving.

THE TAXONOMY OF THIS CHAPTER STOPPED ONE TIER SHORT, and it took a specification from outside the caching literature to show where the missing one sits. On June 12, 2026, two Google engineers published the Open Knowledge Format: organizational knowledge laid down as a git directory of typed markdown concepts — one file per definition, per metric, per procedure — each carrying frontmatter that records who wrote it, who checked it, what it derives from, and when it stops being true.⁴⁶ Nowhere does the spec say the word *cache*. But read it with this chapter's eyes and the whole design is a cache argument: OKF moves intelligence from query time to *authoring* time — the expensive acts of comprehension performed once, under review, then served many times. A governed concept file is *cached understanding*. Part I's founding economics — the cheapest retrieval is the one that never runs — acquires an elder sibling: the cheapest comprehension is the one performed before the question arrived.

Seat it on the placement map. The five layers all cache *at query time*, keyed by the query: they store what the pipeline computed the last time this question arrived, and they inherit every staleness pathology of computing under deadline. Layer five was the deepest of the five — cache the meaning, not the words — and the governed layer is what layer five becomes when the caching happens *before any query exists*: keyed by concept rather than by query, written by an author rather than by a pipeline, and reviewed before it may serve rather than verified after it has. It is not a sixth valve on the pipeline; it sits upstream of the pipeline entirely, in the corpus — which is why the chapter could walk all five valves and never meet it. The proposal-plus-gate architecture governs here too, its fifth appearance in this corpus and its first at authoring time: an extraction agent proposes a concept as a reviewable diff, a domain owner's merge is the gate, and merge *is* the promotion event, recorded as a verification. OKFcRAFT's blunt caveat carries over unchanged: an unreviewed bundle is just chunks wearing a suit — a cache of understanding nobody actually performed.

THE LAYER'S EXPIRY FIELD CORRECTS OUR OLDEST CLOCK, and the correction deserves a lecture slide of its own. The taught invalidation triple begins with a TTL — a *relative* clock, started at write time, set by the operator's guess about volatility. OKF's `stale_after` refuses the relative form: it is an *absolute date*, stamped at write time by the author, who is the one party that actually knows the fact's shelf life — the pricing policy expires with the fiscal year, not *n* seconds after somebody happened to cache a question about it. The spec's stated reason is a systems designer's reason: staleness becomes a plain date comparison, with no reference to when the concept was read — no per-reader arithmetic, no serve-time state, the same verdict from every consumer at the same instant.⁴⁷ Knowledge, in this design, ages like milk, not like wine: the carton bears a date, not a countdown. Notice how cleanly this lands on the principle of the previous section. A TTL encodes the consumer's guess about the world; `stale_after` encodes the author's *knowledge* of it — a fact about the answer, carried by the entry, which is exactly what “invalidate by evidence, not by clock” asked for. Clocks stay demoted to backstops — but a backstop set by the party who knows beats a backstop guessed by the party who serves.

NOW PUT THE FRONT DOOR IN FRONT OF A GOVERNED PIPELINE, because that composition is where this report has something to say that OKFcRAFT does not. The pipeline answers definitional and numeric questions from concepts that carry papers: a trust tier — unverified, machine-confirmed, human-reviewed — *derived*, never stored, from verification events; and, for numbers, an attested computation whose deterministic attester keeps the LLM banished from the verdict. The semantic cache stores that answer and serves it at five milliseconds. If the hit does not carry the tier and the attestation status forward, the cache has performed the trick Part VII taught us to smell in vendor copy — laundering. Accuracy–frequency laundering re-typed a precision as a recall; this is *provenance laundering*, and it is worse: an answer minted from an unverified draft emerges from the cache indistinguishable from one minted from a human-reviewed policy, wearing the borrowed authority of instant, confident delivery. The gauntlet therefore grows a fifth gate — *provenance-tier identity*. The entry stores the IDs of the concepts it derived from (the schema's back-pointer, load-bearing yet again), with the `stale_after` and `status` it saw at mint time; at hit time the gate re-derives tier and lifecycle from the live bundle and refuses to serve if any cited concept has been deprecated, has passed its `stale_after`, or now derives a *lower* tier than the one the answer was minted under. The derived-never-stored discipline applies to caches with double force:

⁴⁷ OKF Specification v0.2, §§5.4–5.5; OKFcRAFT Part III reads the lifecycle fields closely. `status` takes `draft`, `stable`, `deprecated` — deprecated concepts are kept for links and history, retired like a library book moved to the annex, not dropped like a database row.

a tier stored as truth in a cache entry is a tier that can rot — demoted in the bundle on Tuesday, served as gospel on Wednesday. Re-derive on read, always.

Principle 0.5 (Trust must travel through the cache). A cached answer derived from governed knowledge must carry its provenance backpointers, and the tier, status, and expiry of every cited concept must be re-derived from the live bundle at serve time — never trusted from the entry. A cache that strips the papers launders a draft into an authority at five milliseconds. A cache that keeps them gains the layer-two virtue at layer one: a concept that expires or is deprecated kills every cached answer citing it, by ID lookup — exact invalidation, reaching into the governed layer.

WHAT THE GOVERNED LAYER COSTS IS THE HONEST CODA, and the report's discipline requires stating it as an open bet, not a won one. The five layers are priced in compute; the governed layer is priced in the one currency that has killed every knowledge initiative before it: sustained human attention at the merge. The wager — OKFCRAFT states it and declines to settle it — is that agents change the economics of curation: machines patrol the staleness queue, re-checking expired concepts against their sources and drafting refresh diffs, while humans adjudicate only where judgment is owed. As of this writing no production deployment has published a postmortem, and the wager is *open*; this report, which spent Part VII refusing vendor arithmetic, will not pretend otherwise. But note the symmetry with our own economics, because it is the reason this section belongs in a caching monograph at all. The cache's premise is repetition; curation is repetition *noticed in advance* — the head of the distribution moved out of the query-time cache and into the corpus, comprehended once, under review, before anyone asks. Even the scouting problem is already solved by machinery this report owns: the near-miss mine, the band just below τ where the organization's almost-repeated questions concentrate, was the cache's second salary — hard negatives for E , skill candidates for the agent — and it now pays a third: *concept candidates* for the governed layer. The front door, whose whole art is recognizing returning questions, is the natural scout for which understanding deserves to be authored. If the wager pays, the deepest cache tier in the house is not the answer store at all; it is the corpus that quietly made half the answer store unnecessary. Heraclitus, once more: the river moves, but some stones can be quarried, stamped, and dated — and the stamp says who cut them, who checked them, and the day they crumble.

Caching for Agents

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EVERY RESULT IN THIS REPORT SO FAR ASSUMED ONE QUIET PREMISE: that a repeated *question* deserves a repeated *answer*. For chat and RAG the premise mostly holds. For agents it fails at the root, twice. First, an agent’s outputs depend on state the query never mentions: “check my calendar and book the flight” asked twice, verbatim, must execute twice, because the calendar moved. Second — subtler and now measured — the embedding geometry that Part II built our confidence on turns actively *treacherous* for tool use: “check my email” and “send an email” are cosine neighbors and operational opposites. The measured consequence of ignoring both failures is stark: the classical semantic cache, pointed at agentic traffic, achieves a hit rate of *three percent* on the literature’s synthetic agent-task workload — and some of those hits are the dangerous kind.⁴⁸ This part is about what survives: not answer caching, but three progressively deeper changes of the *cacheable unit*.

The Wrong Equivalence, Named Precisely

COSINE SIMILARITY MEASURES TOPICAL KINSHIP; tool dispatch requires *operational* equivalence — same tool, same parameters, same effect — and the two relations disagree exactly where it hurts. Verbs that invert meaning barely move an embedding (check/send, enable/disable, buy/sell); parameters that change everything are semantically invisible (\$100 versus \$100,000 transfers); while genuinely equivalent commands may share no surface at all (“clear my afternoon” and “cancel today’s

⁴⁸ The 3.3% GPTCache figure (on that paper’s synthetic agent-task benchmark), and the check-email/send-email cosine-neighbor diagnosis: *Structured Intent Canonicalization with Few-Shot Learning*, arXiv:2602.18922 (2026). The agents-course bonus-cost chapter’s practitioner ladder agrees from the field: 5–15% hit rates on tool-call traffic, the lowest of any workload class.

meetings after 1pm”). In Part II’s language: for agents, the bears in a cap are not near-synonyms of your query — they are its *siblings, rivals, and negations*, and the within-cap adjudication problem that Part III solved statistically for answers becomes, for actions, unforgivable to solve statistically at all. A ninety-eight-percent-confident “send” when the user said “check” is not a quality degradation; it is an incident report.

Principle 0.6 (For actions, equivalence must be decided, not scored). An answer served at $p = 0.98$ is a good bet; an action dispatched at $p = 0.98$ is a two-percent incident rate. Caching for agents therefore abandons the similarity threshold as the deciding instrument and rebuilds equivalence as something checkable: a canonical form, a state key, a schema match. Where Part III priced the bet, this part’s systems refuse it.

Canonicalize the Intent, Key on the Skeleton

THE FIRST SURVIVING DESIGN REPLACES THE EMBEDDING KEY WITH A STRUCTURED ONE. Decompose each request into its journalist’s skeleton — who, what, when, where, why, how, how much — and observe the asymmetry the decomposition reveals: the *what* and *where* (the operation and its target system) define the cacheable route, while the parameters — amounts, dates, recipients — must be kept *out* of the key entirely and re-bound at execution. “Transfer \$100 to Sarah” and “send \$500 to Ahmed” share a route (funds-transfer via the banking tool) and share nothing executable; the cache should recognize the route instantly and re-parameterize it freshly, every time. The keying discipline is enforced not by a threshold but by a small classifier trained to canonicalize consistently — and the measured comparison is almost embarrassing to the giants: a few-shot fine-tuned small model canonicalizes at ninety-one percent accuracy in single-digit milliseconds, against sixty-nine percent for a twenty-billion-parameter model reasoning at three seconds. Consistency, not intelligence, is the job.⁴⁹

⁴⁹ W5H2 canonicalization, arXiv:2602.18922: (What, Where) as the cache key with parameters excluded; SetFit-class few-shot classifier at 91.1%/2–5ms vs 68.8%/3.4s for a 20B general model; a five-tier cascade resolving 85% of interactions locally.

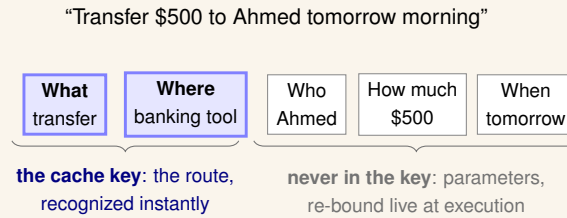


Figure 17: The canonicalization discipline: key on the skeleton, re-bind the flesh. “Transfer \$100 to Sarah” and “send \$500 to Ahmed” share a route and share nothing executable — the cache recognizes the first and must never reuse the second. A small classifier does this at 91% in single-digit milliseconds; consistency, not intelligence, is the job.

PUSH CANONICALIZATION TO ITS LIMIT AND SOMETHING WONDERFUL HAPPENS: THE 1996 THEOREM COMES HOME. Part I told the lineage as a fall — decidable predicate containment traded for a geometric bet. In domains with schemas, the trade reverses. The analytics world now canonicalizes both SQL and natural-language questions into typed *intent signatures* — measures, dimensions, filters, time windows — validated against the warehouse schema, with roll-up and filter-down *derivation* between cached signatures: a cached quarterly aggregate can *prove* it answers a monthly question the way a 1996 cache proved containment. Result: eighty-two percent hit rates with *zero* spurious matches — recall three times higher than text keying, at perfect precision.⁵⁰ The motif deserves its sentence in the lecture: *wherever a schema exists, stop betting and go back to proving*.

Cache the Plan, Not the Answer

THE SECOND SURVIVING DESIGN MOVES THE CACHEABLE UNIT UP A LEVEL OF ABSTRACTION. What repeats across an agent’s episodes is rarely the answer and often the *shape of the work*: fetch, then filter, then summarize, then notify — the same skeleton over ever-fresh state. So cache *plan templates*: extract the structure from completed successful episodes, strip the bindings, store the skeleton keyed by task type; on a repeat, a lightweight model adapts the template to today’s parameters while every tool call executes live against the current world. Measured on agent benchmarks: half the cost, a quarter less latency, quality preserved — because nothing stale ever reaches the user; only the *thinking about how to proceed* was reused.⁵¹

The rhyme with this corpus is strong enough that the lecture can lean on it: a cached plan is a *proto-skill* — procedural knowledge extracted from episodes, reused across instances — sitting exactly one maturity rung below SKILLCRAFT’s promoted procedures and MEMORYCRAFT’s session-to-skill consolidation pipeline. The cache, here, is the nursery

⁵⁰ OLAP intent-signature caching, arXiv:2602.19811 (2026): canonical OLAP Intent Signatures, strict schema validation, confidence-gated NL acceptance, derivation-aware reuse; 82% vs 28% (text-keyed) on TPC-DS-class workloads, zero spurious hits.

⁵¹ Agentic Plan Caching, Zhang, Wornow, Wan, Olukotun, arXiv:2506.14852, NeurIPS 2025: plan-template extraction with keyword keying and lightweight adaptation; −50.3% cost, −27.3% latency at maintained quality.

Sibling-rhyme ledger: layer five closes the loop with the procedural tier of the memory taxonomy. Write → consolidate → promote: the agent's cache is where its habits audition.

of procedural memory: plans that keep earning hits are plans begging to be promoted into skills, and the near-miss mining that the taught chapter used to harvest fine-tuning pairs becomes, for agents, a *skill-candidate* miner.

Key on the State, When State Is the Point

THE THIRD DESIGN SOLVES THE STALENESS PROBLEM BY DISSOLVING IT: make the *state* part of the key. In reinforcement-learning post-training — where the same agent replays thousands of near-identical episodes and tool latency dominates wall-clock — caching tool *outputs* naively is wrong for exactly the calendar-moved reason. The repair: key each tool result on the *entire history of calls that produced the current environment state*, organized as a tree with longest-prefix matching. Two episodes that have made identical calls so far are, by construction, in identical states; their next identical call may be served from cache with a guarantee, not a bet. Hit rates up to seventy percent, median tool execution up to seven-fold faster, and — the number that matters — zero reward degradation, because state equivalence was provable.⁵² The design generalizes beyond RL to any agent whose tools are deterministic given history — and it is the purest expression of this part's law: when you cannot prove equivalence of *meanings*, engineer the system until you can prove equivalence of *states*.

⁵² TVCache, Vijaya Kumar et al. (Cornell), arXiv:2602.10986 (2026): tool-call caching for agentic RL keyed on full call-history prefixes; up to 70% hit rates and up to 6.9× faster median tool execution (ceilings, not typicals), no reward change.

The agent-caching ladder, on one card

Never: answer-cache an action by cosine — 3% hits and the misses are incidents. *Canonicalize*: structured intent keys, (What, Where) route with parameters re-bound live — and full decidable signatures wherever a schema exists. *Plan-cache*: reuse the skeleton of the work, execute everything live; promote persistent winners toward the skill library. *State-key*: when tools are deterministic given history, key on the history and serve with proof. Each rung trades generality for certainty; agents, unlike chatbots, must climb.

STEP BACK. Part V ended by calling layer five the deepest bet on repetition; this part sharpened it: for agents, what repeats is never the world — it is *structure*: routes, plans, histories. Cache the structure, rebind the world. And notice what the whole part quietly demonstrated: every successful agent-caching design abandoned the similarity threshold in

favor of something checkable — a canonical form, a schema proof, a state key. The one cache family that cannot afford to be wrong simply left the betting business. There is a lesson in that for the highest-stakes corners of the answer cache too, and the Interlude, next, supplies the adversarial reason the lesson generalizes.

Interlude: The Adversary at the Door

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EVERY SECURITY CHAPTER IN THIS CORPUS BEGINS THE SAME WAY, because the pattern never stops being true: the capability we celebrate is the attack surface we ship. MEMORYCRAFT said it of self-editing memory; SKILLCRAFT of self-improving procedures. For the semantic cache the pattern arrives with a theorem attached, and the theorem is the reason this is an interlude rather than a section: it cuts across every part of this report at once. The cache’s entire value — serving neighbors, not just twins — rests on the *locality* of its keys: near meanings, near vectors, shared hits. In January 2026 it was proved, and then demonstrated against production stacks, that this same locality is a cryptographic defect. A hash function is secure because near inputs land far apart; a semantic key is *useful* because near inputs land together. You cannot have both properties in one keyspace. Hit rate and collision resistance are one quantity with two signs.⁵³

The Bear Suit, Fitted

PART II PROMISED THIS ANATOMY: THE CAP THAT ADMITS BEARS CAN BE ENTERED DELIBERATELY. The attack is exactly a tailoring problem. The adversary picks a victim query — “what is the wire transfer limit?” — and, by gradient search over token substitutions, crafts a prompt that a human reads as unrelated noise but whose *embedding* lands inside the victim’s polar cap: a bear suit, stitched in embedding space. Two uses

⁵³ Zhang, Liu, Xie, Huang, She (HKUST), *From Similarity to Vulnerability: Key Collision Attacks on LLM Semantic Caching*, arXiv:2601.23088, ICML 2026. Lemma 3.1 formalizes the tradeoff: raising hit probability provably raises the false-positive collision risk available to an adversary. The demonstrations, not just the lemma, are the news.

The financial-agent case study in the paper executes unintended orders through a collided cache entry. Read alongside Part VI's principle — actions must be decided, not scored — the two parts of this report were written by the same threat model.

follow, one loud, one quiet. The loud one is *poisoning by collision*: the adversary asks their crafted prompt *first*, the system dutifully generates and caches an answer under a key that sits in the victim's cap — optionally an answer the adversary has steered — and the legitimate user's next question collides with it and is served the adversary's entry at five milliseconds, with the cache's full confidence. The quiet one is the *timing side channel*: a hit returns in milliseconds, a miss in seconds, so an adversary can probe a shared cache with crafted neighbors and learn, from latency alone, *what other tenants have been asking*. The measured success rates against reference stacks — including semantic-cache architectures on the major clouds — run from the high seventies to ninety-one percent, and the agent-facing variant hijacks cached *tool invocations*: Part VI's incident report, delivered by a stranger. Concentration of measure, which guaranteed that nothing wanders into the cap by chance, is not contradicted — it is *recruited*. The attack works precisely because arrival in the cap is always a caused event: the geometry authenticates that something was placed, never *who* placed it or *why*. The cap has no passport control.

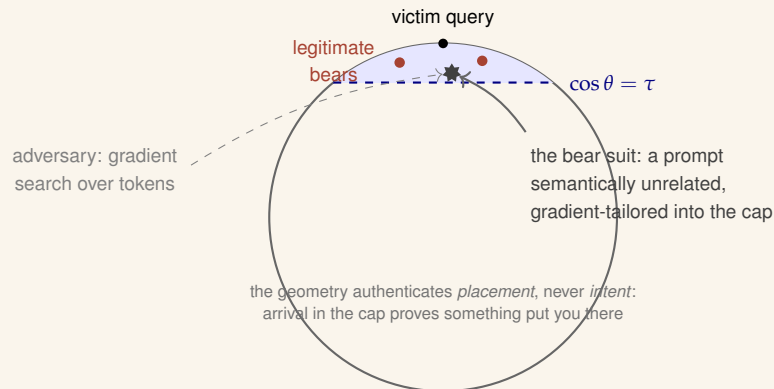


Figure 18: The key-collision attack in habitat terms. The cap admits only the placed — but placement can be purchased by gradient search as well as earned by training. The crafted point wears cosine above τ and serves the adversary's cached content to the victim's queries. Locality is collision risk; scope is the defense that does not bet.

Poisoning Without an Adversary

THE AMPLIFICATION HAZARD DOES NOT EVEN REQUIRE MALICE. A cache is a broadcast tower for whatever it stores: one hallucinated answer, cached, becomes that hallucination served to every neighbor of the unlucky key until invalidation — the taught corpus's audit line (“generated at 14:32, served at 14:45 — poison”) scaled from one user to a population. The practitioner record now includes the full folklore: the phantom policy answer served for weeks; the embedding-model

upgrade that silently re-keyed a cache and corrupted its neighborhoods; the cross-tenant leak where user B received user A's order status because nobody scoped the keyspace. Add the deliberate attacker and the write path becomes the security boundary: *anything that can cause a cache write with attacker-influenced content is a publishing pipeline to every future colliding query*. The taught write-side salience gate was quality control; it must now also be treated as what MEMORYCRAFT's governor was: a trust boundary with provenance checks, because a cache entry is a memory that many users will be forced to share.

Defenses, Priced Honestly

EVERY PUBLISHED DEFENSE BUYS SAFETY WITH HIT RATE, exactly as the theorem predicts; the craft is choosing which coin to pay. *Key salting* perturbs the embedding space per deployment, raising the attacker's search cost — and blurring legitimate neighborhoods. *Perplexity screening* rejects the gibberish-looking adversarial prompts — until the attacker constrains fluency, at higher search cost; a useful ratchet, not a wall. *Centroid matching* replaces single-entry nearest-neighbor with cluster-level matching — match against the community's center, not the nearest point — so a single implanted outlier cannot sit as anyone's nearest neighbor; the measured version cuts adversarial success roughly fourfold, at the price of coarser matching.⁵⁴ *Verification* (Part III's family) is a defense wearing its other hat: a synchronous check that the cached answer actually answers the query defeats most collision-served nonsense, at verification latency. And *structural scoping* — the strongest of all — shrinks the blast radius by construction: per-user or per-tenant keyspaces mean the adversary can only collide with *themselves*. The timing channel closes the same way.

⁵⁴ SAFE-CACHE, Scientific Reports 16:5936 (2026): Louvain-community centroid matching with purity screening and adversarially fine-tuned refinement; 52.8% → 14.3% attack success against the GPTCache baseline.

defense	what it buys	what it costs
key salting	raises attacker search cost	blurs honest neighborhoods
perplexity screen	rejects gibberish suits	ratchet, not a wall
centroid matching	no single-point nearest	coarser matching
verification	catches served nonsense	latency + judge cost
scoping	removes the target	hit rate (smaller populations)

Figure 19: Every defense pays in the currency the theorem names — and only scoping changes the game rather than the odds. The premium belongs on the economics invoice *before* the build decision.

Principle 0.7 (Scope is the only defense that does not bet). Salting, screening, centroids, and verification all raise the attacker’s cost; partitioning *removes the target*. A cache entry should be reachable exactly by the population entitled to its answer — per-tenant always, per-user wherever personalization touches content, shared only for anonymous public knowledge, and then behind verification. This is RETRIEVAL-CRAFT’s cabinet of drawers and MEMORYCRAFT’s partition interlude, arrived at their sharpest consequence: in 2026, *a shared, unscoped semantic cache is not a design smell — it is a published exploit target, and shipping one is malpractice*.

THE VENDORS, TO THEIR CREDIT, HEARD THE SAME THUNDER: the scoping features that appeared across managed caches and gateways in 2025–26 — per-user and per-session key partitions, attribute filters, subscription-keyed cache variance — are this interlude in product form, and Part VIII will read their documentation with that lens. What no product yet ships is the honest dashboard this interlude demands, so the requirement lands on the builder and the capstone team: alongside hit rate and demonstrated error, report *scope coverage* (what fraction of entries are reachable beyond their author) and *collision exposure* (whether any shared tier serves consequential answers unverified). Two numbers, one glance, and the difference between a front door and an unlocked one.

THE INTERLUDE’S RESIDUE, CARRIED FORWARD: the economics of Part VII must now include the cost of defense — scoped caches have lower hit rates, verified hits have higher latency, and both subtractions belong in the break-even arithmetic *before* the build decision, not after the incident. The bet the semantic cache places was priced by Part III for honest traffic; the adversary’s premium goes on the same invoice.

The Economics: After Prefix Caching Ate the Easy Win

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IN 2023 THE SEMANTIC CACHE SOLD ITSELF: model calls were expensive, every repeated question was pure waste, and a vector store in front of the API was found money. That pitch is dead, and honesty about *who killed it* is the beginning of the real analysis. The killers were the providers themselves. Between 2024 and 2026 all three major vendors shipped *prefix caching* — exact, token-level, zero-correctness-risk reuse of repeated context — and converged on roughly ninety-percent discounts for cached input tokens, in OpenAI’s and Gemini’s case activating *automatically*, with no infrastructure, no threshold, and no way to be wrong.⁵⁵ The repeated *context* — system prompts, tool schemas, shared documents, the bulk of most bills’ input side — is now discounted at the source. What the 2023 pitch counted as the semantic cache’s savings was mostly this, and it has been claimed by someone who takes none of the semantic cache’s risks. GPTRCache did not die of a rival cache; it died of a price change.

What Is Left to Save

STRIP AWAY WHAT PREFIX CACHING NOW COVERS, AND THE SEMANTIC CACHE’S HONEST KERNEL HAS EXACTLY THREE TERMS. *Output tokens*: a prefix-cached call still generates its answer at full price, and output tokens run three to six times the input rate — on chat-shaped workloads

⁵⁵ August-2026 parameters, from vendor documentation (mechanically verifiable pricing): Anthropic — explicit `cache_control`, 5-minute and 1-hour tiers, writes at $1.25\times/2\times$, reads at $0.1\times$; OpenAI — automatic at ≥ 1024 -token prefixes, model-dependent discounts 50–90%, extended retention to 24h on current flagships; Gemini — implicit caching by default plus explicit cached contents billed as storage per MTok-hour, cached tokens $\sim 90\%$ off.

The sequencing doctrine every credible 2026 practitioner account converges on, quotable as-is in the classroom: *prefix caching is the default; the semantic cache is a scalpel*. Turn on the free, riskless discount first — one financial-services team found it captured sixty percent of the total win at zero engineering — then deploy the scalpel only on the identifiably repetitive slices that remain.

they are the majority of spend; a semantic hit is the only cache that avoids them entirely. *The whole pipeline*: for RAG, a hit skips embedding, search, rerank, and assembly — costs no provider discount touches. *Latency*: seconds of generation become milliseconds of lookup, which is not a discount but a *product feature* — and in voice interfaces, where a two-second pause is a failed conversation, it is the entire reason the niche’s healthiest businesses exist.

The Break-Even, Derived at Last

THE GAP ANALYSIS SCRIBBLED A BREAK-EVEN FORMULA AND NEVER TAUGHT IT; HERE IT IS, DERIVED, WITH ITS TEETH SHOWING. Let N be daily query volume, H the hit rate, C_{full} the marginal cost of a full pipeline call *after prefix discounts* (input remainder + all output tokens + retrieval infrastructure), C_{hit} the marginal cost of serving a hit (embedding, search, verification), and C_{ops} the daily fixed cost of the cache itself — vector store, monitoring, the amortized engineering and the weekly graded samples the error contract demands. The daily net is

$$\Delta = N \cdot H \cdot (C_{\text{full}} - C_{\text{hit}}) - C_{\text{ops}},$$

and the cache pays if and only if the hit rate clears

$$H^* = \frac{C_{\text{ops}}}{N (C_{\text{full}} - C_{\text{hit}})}.$$

Read the denominator with 2026 eyes, because both of its factors have been shrinking. C_{full} falls every quarter — flagships get cheaper, and the nano-tier models run three orders of magnitude below them — while C_{hit} is roughly constant physics: an embedding call, an index probe, perhaps a verification. *Falling model prices raise the bar the cache must clear*.

Two worked verdicts, one from each side of the bar. A support assistant on a flagship model: $N = 50,000/\text{day}$, $C_{\text{full}} \approx \0.02 (output-heavy), $C_{\text{hit}} \approx \$0.0004$, $C_{\text{ops}} \approx \$40/\text{day}$ all-in — $H^* \approx 4\%$, and the observed 40–60% FAQ hit rate clears it by an order of magnitude: build. The same assistant migrated to a nano-tier model: $C_{\text{full}} \approx \0.0006 , and suddenly $H^* \approx 400\%$ — unreachable at any hit rate: *the cache loses money on every configuration* — all-in, with C_{ops} amortized per query, its embedding and operations cost more than the inference it avoids. The correct engineering decision is the one our taught viability gate always implied and this formula now prices: do not build.

RETRIEVALCRAFT Part IX’s ledger entry said it in the two-column voice: the cache posts to the cost column, not the quality column — and *nothing degrades without it; that is the point of the entry*. The break-even formula is that verdict with numbers in it. The latency exception survives the arithmetic: when milliseconds are the product, Δ is measured in conversions, not tokens, and voice keeps building caches that lose money on tokens and earn it back on answers-before-the-pause.

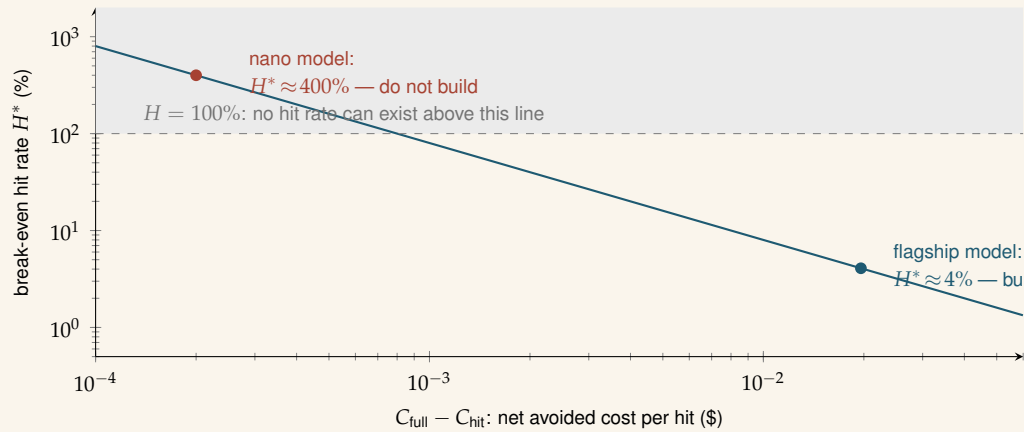


Figure 20: The break-even frontier, $H^* = C_{\text{ops}}/N(C_{\text{full}} - C_{\text{hit}})$, drawn for the worked example ($N = 50,000/\text{day}$, $C_{\text{ops}} = \$40/\text{day}$). Cheap models push the required hit rate off the top of the chart: falling inference prices raise the bar the cache must clear. Read it as a phase diagram: the curve is the phase boundary between build and do-not-build, and falling inference prices move every deployment steadily toward the disordered phase. The shaded region is not pessimism; it is arithmetic.

TWO SUBTRACTIONS BELONG ON THE INVOICE BEFORE THE VERDICT, AND BOTH COME FROM EARLIER PARTS. The *defense premium*: the Interlude’s scoping shrinks shareable populations — per-user partitions can halve an open cache’s hit rate — and verification adds latency and judge cost to every guarded hit; H and C_{hit} must be quoted *after* these, not before, because the alternative is discovering the premium in an incident report. And the *measurement tax*: the error contract of Part III runs on labelled exploration — a standing percentage of hits double-served and compared — which is a real, recurring line in C_{ops} and the cheapest insurance in this entire report.

Reading Vendor Arithmetic

NOW THE HYGIENE SECTION, BECAUSE THE NUMBERS YOU WILL MEET IN THE WILD ARE LAUNDERED. The recurring trick deserves a name and a reflex: *accuracy–frequency laundering*. A vendor measures that ninety-five percent of served hits were correct — a precision, creditable — and the marketing funnel re-types it as “95% hit rate,” a recall, implying near-total traffic coverage. The honest field numbers, assembled across practitioner reports and one refreshingly candid vendor’s own aggregate: FAQ and support traffic 40–60%; classification 50–70%; RAG Q&A 15–25%; open-ended chat 10–20%; agentic traffic 5–15%; blended reality near twenty-five percent — against calculator marketing built on 85% assumptions.⁵⁶ RETRIEVALCRAFT’s one-bit rule applies with full force: a vendor-run benchmark proves the system is in the competitive range, and one bit is all it proves. The two questions that X-ray any

⁵⁶ The practitioner ladder and the laundering diagnosis: the “95% Hit Rate Myth” analysis (dev.to, April 2026, author affiliated with cache tooling — read critically); the ~20% figure from Portkey’s published early production tests (18–60% by workload); the threshold-versus-false-positive frontier from Respan’s production guide (0.93–0.95 similarity $\Rightarrow$ 25–40% hits with 3–7% of hits wrong). All flagged [P] in the source ledger; the convergence across independent sources, not any single number, is the evidence.

cache claim: *is that number a precision or a recall? And measured on whose workload?*

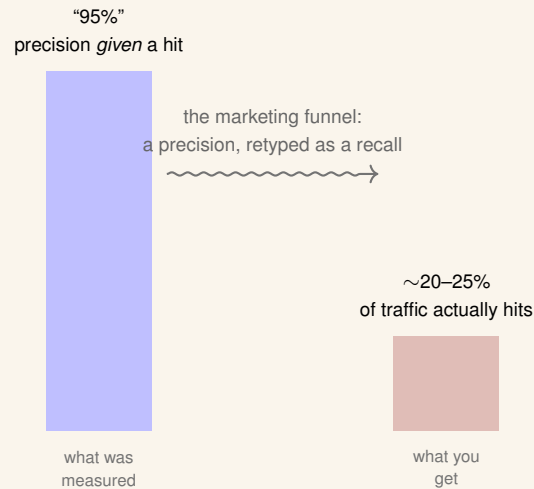


Figure 21: Accuracy–frequency laundering, the recurring trick of cache marketing. Both numbers are real; only one is the one implied. The two questions that X-ray any claim: precision or recall? Measured on whose workload?

The economics, on one card

Prefix caching first — free, riskless, already ninety percent of input-side savings. Then the formula: $H^* = C_{\text{ops}} / N(C_{\text{full}} - C_{\text{hit}})$, with C_{full} taken *after* prefix discounts, H and C_{hit} taken *after* scoping and verification, and the viability gate (one-in-ten at 0.85) as the free pre-measurement of whether H can exist at all. Cheap models raise H^* ; latency products ignore it; and “do not build” is a success of the analysis, not a failure of nerve.

STEP BACK. The 2023 story was “caching saves money”; the 2026 story is a division of labor: the *provider* caches your repeated context, the *scalpel* caches your repeated questions where they are dense, safe, and scoped — and the spreadsheet, not the enthusiasm, decides where the scalpel cuts. What remains is to see who is actually selling the scalpel, who quietly stopped, and what their documentation admits when read closely. That tour is next.

In the Wild

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READ AN INDUSTRY NOT BY ITS ANNOUNCEMENTS BUT BY ITS REVEALED PREFERENCES, and the semantic-cache market of August 2026 tells a story more honest than any single vendor’s copy: one database company betting its agent strategy on it, one hyperscaler shipping it with a warning label, one gateway pricing it into the enterprise tier, two infrastructure companies declining to ship it at all, one early vendor pivoting away, and the field’s founding open-source project abandoned. Every stance is evidence. This part walks the market with the one-bit rule in hand — a vendor claim proves membership in the competitive range, nothing more — and extracts what the products teach that the papers cannot.

The Believers

REDIS IS THE MARKET’S COMMITTED BULL, and its trajectory is the most instructive artifact on the shelf. LangCache — a fully managed semantic cache, REST-fronted, embeddings generated service-side — entered public preview in September 2025 and remains in preview a year later; in May 2026 it was folded into “Redis Iris,” a bundled context engine for agents, as the save-on-repeated-questions organ.⁵⁷ Three lessons ride on it. The scoping features are the Interlude wearing a product UI — the market absorbed the multi-tenancy lesson before most curricula did. The purpose-built embedder line (Part IV’s story: v1 struggling to beat off-the-shelf ModernBERT-class models, v3 retrained small on eight million labelled pairs) shows a vendor discovering in public that cache

⁵⁷ Redis LangCache: public preview 2025-09-04, still preview-labelled in Redis Cloud docs mid-2026; bundled into Redis Iris May 2026. Scoping by user, app, session, and attribute filters — the Interlude in product form. Preview pricing ~\$1.50/M input tokens processed; the calculator’s headline savings assume an 85% hit rate. [V]

matching is a specialist problem. And the pricing — per input token *processed*, hit or miss — quietly relocates the break-even of Part VII: with a managed cache you pay C_{hit} on every miss too.

MICROSOFT SHIPS THE MOST ENTERPRISE-REAL VERSION — semantic-cache policies in Azure API Management, provider-agnostic, Redis-backed — and earns this report’s respect for a sentence its own documentation contains: the semantic cache “may surface incorrect, outdated, or unsafe responses... evaluate carefully and implement safeguards.” A hyperscaler selling the product and warning against it in the same document is not confusion; it is the five dangers, notarized.⁵⁸ Kong ships the most mature gateway plugin — pgvector and Redis backends, streaming support — now gated to its enterprise tier: the market pricing semantic caching as an *enterprise* concern is itself a datum about where the value concentrates. And Portkey deserves the candor prize: semantic mode enterprise-only, cosine default 0.95 — and a published figure from its own early production tests of roughly *twenty percent* hit rates (18–60% by workload), the most honest number any vendor has volunteered and the anchor of Part VII’s ladder.

⁵⁸ Azure APIM `llm-semantic-cache-lookup` policy family: external Redis-compatible store, vary-by partition expressions, ignore-system-messages; the quoted warning is verbatim from Microsoft’s own policy documentation. [V, self-aware]

The Skeptics, the Pivot, and the Grave

NOW THE NEGATIVE SPACE, WHICH TEACHES MORE. *Cloudflare* — owner of more caching DNA than any company alive — ships exact-match caching only in its AI gateway, with semantic “planned” in its documentation year after year: a revealed preference from the people best positioned to know what approximate matching costs at edge scale. *Helicone*: exact only, by design, documented plainly. *LiteLLM* ships semantic backends in name, but its issue tracker tells the operational truth: a March 2026 report documenting the Qdrant semantic path “completely non-functional” — startup crash, request crash, silent write failures — was closed *not planned*, joining a lineage of similar reports.⁵⁹ The lesson for builders is blunt: gateway checkbox semantic caching is, in 2026, more often shelfware than system, and the taught architecture — own your cache, own your policy — was the right instinct all along. *Momento* supplies the pivot: an early semantic-cache vendor that walked to the other altitude entirely, rebuilding around KV-cache offload for inference engines — concluding, with its balance sheet, that the durable money is in caching *computation*, not answers. And *GPTCache* lies where Part I left it: last release August 2024, integrations bit-rotted, survived by its own citation. The field’s reference architecture is now a reference

⁵⁹ LiteLLM issues #23441 (2026-03, closed not-planned), #19163, #14889 — the semantic-cache paths rotting in the open. [P: the issue tracker, not the docs, is the document of record.]

document.

The Tower of Thresholds

ONE INCOHERENCE SPANS THE ENTIRE MARKET AND BELONGS IN THE LECTURE AS A LIVE HAZARD. There is no agreement on what the threshold dial *means*. Portkey: cosine *similarity*, default 0.95, higher is stricter. Azure: a *distance* score, recommended 0.05, higher is *looser*, with a warning beyond 0.2. Kong: distance again, default 0.2. The AWS reference architecture: similarity, 0.75. A student who copies “threshold 0.2” from a Kong config into a Portkey deployment has not loosened the cache slightly — they have *inverted its meaning*, admitting nearly everything the index returns.

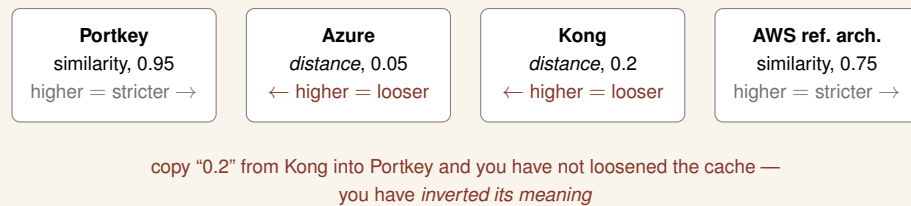


Figure 22: The tower of thresholds: four products, two units, opposite directions of strictness — and every one of them a global static dial, the abstraction the literature has already retired. Check the sign before you turn.

Check the sign of the dial

Before touching any product’s threshold: is it similarity or distance? What metric, over normalized vectors or raw? Which direction is stricter? The industry answers all three differently, sometimes within one vendor’s product line — and Part III’s deeper verdict stands behind the confusion: every one of these dials is a global static threshold, the abstraction the literature has already retired. The products ship rung one of the ladder; the papers live on rung three. That gap — shipped practice trailing published knowledge by two rungs — is the single most exploitable fact in this part, for a capstone or for a product.

STEP BACK, AND READ THE MARKET’S VERDICT AS A WHOLE. Semantic caching in 2026 is neither the 2023 miracle nor a failed idea: it is a *specialist instrument consolidating into the hands of parties with dense, repetitive, scoped traffic* — managed platforms, enterprise gateways, voice products — while general-purpose checkbox implementations rot and the infrastructure elite abstain. The market has, in its own currency,

reproduced this report's argument: the front door pays where repetition is real, judgment is priced, and scope is enforced — and nowhere else. What remains is to fold everything into the pattern language and close the rhymes. The last part follows.

A Pattern Language for Caching, and a Synthesis

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A REPORT EARNS ITS KEEP WHEN ITS ARGUMENTS COMPRESS INTO SHAPES A TIRED ENGINEER CAN REACH FOR AT A WHITEBOARD. Ten patterns, then, in the house format — intent, forces, structure, when to use — each carrying the part that derived it; four anti-patterns, each an operation without a policy; the rhyme ledger that binds this volume to its siblings; and the closing picture.

The Patterns

Front Door. *Intent:* answer returning questions before the expensive machinery wakes. *Forces:* repetition is real but head-heavy; the tail must miss cheaply. *Structure:* exact-hash tier, then semantic tier, then the pipeline; write-back on generation. *Use when:* the viability gate passes — one in ten sampled queries matching at 0.85, or do not build. (Parts I, VII; taught.)

Canonical Key. *Intent:* cache the meaning, not the wording. *Forces:* raw strings fragment the key space; transformation already normalizes. *Structure:* key on the canonised query — referents resolved, frame hashed, volatile message embedded; for agents, the (What, Where) route with parameters excluded and re-bound live. *Use when:* always; it is the coupling doctrine. (Parts I, IV, VI.)

Two-Tier Threshold. *Intent:* extract value from the near-miss band without serving it. *Forces:* the band between τ_{partial} and τ_{full} is too risky to serve, too rich to discard. *Structure:* serve above τ_{full} ; below it, use the neighbor as a *hint* — enrich the query, seed retrieval. *Use when:* any full-pipeline cache. (Taught; RETRIEVALCRAFT’s house refinement.)

Error Contract. *Intent:* replace the threshold you chose with the error you can afford. *Forces:* correct and incorrect hits overlap per-entry; a global τ averages neighborhoods. *Structure:* stated budget δ ; per-entry decision curves learned from exploration labels; demonstrated error reported weekly. *Use when:* traffic is dense enough to feed the curves; stakes justify the exploration tax. (Part III.)

Verified Hit. *Intent:* convert evidence into assurance before or after serving. *Forces:* similarity is evidence, not proof; latency budgets differ by stake. *Structure:* asynchronous judge on the near-miss band, promoting blessed pairs; synchronous check where wrongness is unaffordable. *Use when:* regulated answers, voice products, any shared tier. (Parts III, Interlude.)

Scoped Cache. *Intent:* make the leak and the collision structurally impossible. *Forces:* locality is collision risk; sharing is hit rate. *Structure:* per-tenant always, per-user where content personalizes; scope key inside the cache key, not in an application check. *Use when:* always — the unscoped variant is the first anti-pattern. (Interlude; RETRIEVALCRAFT’s cabinet.)

Version Pin. *Intent:* an entry minted under one geometry must die with it. *Forces:* embedder and index upgrades silently re-key space; corrupted neighborhoods look like drift. *Structure:* embedder and index versions in the entry schema; version mismatch is an unconditional miss; re-embed to migrate. *Use when:* always; invalidation is tied to index versions, full stop. (Taught; Part IV.)

Evidence Gauntlet. *Intent:* freshness checked against the world, not the clock. *Forces:* TTLs guess; document triggers over-kill; meaning changes under stable identities. *Structure:* serial gates at hit time — similarity, evidence overlap, chunk-version identity, lexical support — with TTLs demoted to backstops. *Use when:* layer-one answer caches over living corpora. (Part V.)

Plan Cache. *Intent:* reuse the shape of the work when the world will

not repeat. *Forces*: agent state staleness; structure repeats when answers cannot. *Structure*: extract skeletons from successful episodes; adapt bindings live; promote persistent winners toward the skill library. *Use when*: agentic workloads — where it is the *only* safe answer-adjacent cache. (Part VI.)

Near-Miss Mine. *Intent*: the cache as instrument, not only as saver. *Forces*: the just-below- τ band concentrates exactly the pairs the encoder confuses. *Structure*: log the band; cluster it; feed it back as hard negatives, emerging-topic alerts, and skill candidates. *Use when*: always — it is the cache's second salary. (Taught; Parts IV, VI.)

The Anti-Patterns

EVERY ANTI-PATTERN BELOW IS AN OPERATION WITHOUT A POLICY, the house diagnosis in its fourth volume. **Global Threshold**: one scalar adjudicating every neighborhood — the abstraction the literature retired; a serve operation without a decision policy. **Unscoped Shared Cache**: every tenant in one keyspace — a published exploit target; a share operation without an access policy. **Hit-Rate Worship**: optimizing the one metric that rises as answers get worse — “hit rate is the dangerous metric; watch the error, or the hit rate will watch you”; a measurement operation without a truth policy. **Checkbox Cache**: enabling a gateway's semantic flag and calling it architecture — unowned thresholds with incoherent semantics atop rotting code paths; a deployment operation without an ownership policy.

The Rhyme Ledger

THIS VOLUME CLOSES ITS ACCOUNTS WITH ITS SIBLINGS the way the house demands: notation reused character-for-character, doctrines cited, one table.

This volume	Sibling	The rhyme
$VoS = p\,g - (1 - p)L - \kappa$	SKILLCRAFT VoL; MEMORYCRAFT VoR	one law, three doors; $(1 - p)L$ is the cache's signature term
Error Contract (proposal + gate) Scoped Cache	GEPA validator; skill governor; memory governor RETRIEVALCRAFT cabinet; MEMORYCRAFT partition interlude	fourth instance of the course's one architecture scope as access control over the operations; structure beats checks
Two altitudes (answers vs KV)	MEMORYCRAFT message-level vs KV-level eviction	the ledgers never mix; third appearance of the commandment
Replacement canon, amended	MEMORYCRAFT cache canon (Bélady, LRU/LFU/ARC)	approximate hits break the ground rules; the canon holds one altitude up, not both
Aperture doctrine, promoted	RETRIEVALCRAFT Part II; FT-course Lévy note; LLM-course geometry week	one theorem, three courses, now one canonical statement — and its three silences named
Near-Miss Mine	taught chapter's diagnostic mirror; SKILLCRAFT promotion pipeline	the cache as nursery: hard negatives for E , skills for the agent
Governed layer; trust through the cache	OKFCRAFT tiers (derived, never stored); stale_after as authored absolute expiry	cached understanding above layer five; the hit carries its papers; back-pointer invalidation reaching into the bundle (rev. Aug. 2026)

Synthesis: The Bet, Priced

NOW STAND ALL THE WAY BACK. Context engineering, this corpus's parent discipline, is the craft of deciding what the model sees; its five sides gave the agent action, knowledge, competence, self-improvement, and a past. The cache is not a sixth side. It is something quieter and stranger: *the decision not to consult the model at all* — the only organ in the entire architecture whose success is measured in forward passes that never happen. Every other craft in this family optimizes what enters the context window; this one optimizes the cases where no window opens.

The report's argument, compressed to its spine: a cache is a bet that the past will repeat. *The geometry makes the bet possible* — concentration annihilates chance, so every hit is a caused event; nothing wanders into the cap, and the bears are always somebody's doing. *The decision theory makes it honest* — within the cap the theorem is silent, correct and incorrect hits overlap entry by entry, and the threshold must grow from a number into a policy with a stated, demonstrated error. *The adversary makes it dangerous* — locality is collision risk by theorem, so scope is not hygiene but survival. *The economics decide whether to place it* — after the providers claimed the repeated context, the surviving stake is output tokens and seconds, it clears the bar only where repetition is dense and honest, and “do not build” is a verdict the analysis is proud to return. And at the highest stakes — actions, schemas, states — the mature move is the one the agent literature already made: *stop betting and go back to proving*.

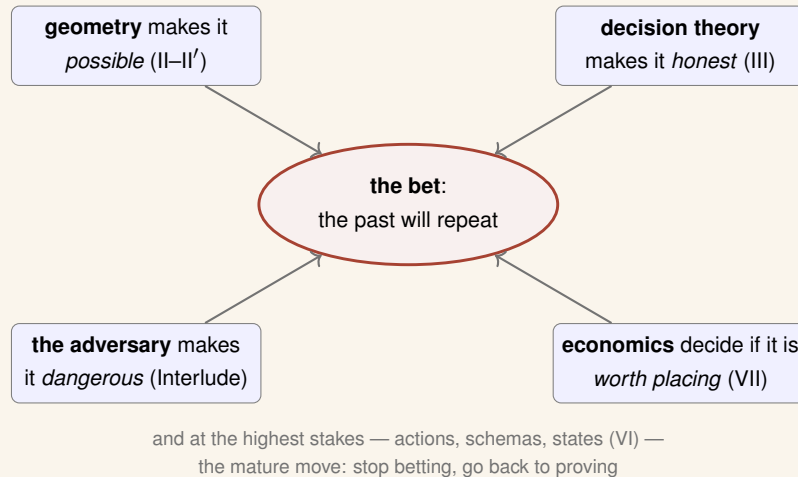


Figure 23: The report on one page: four verdicts on one bet. The implementation parts (IV–V) are the machinery for placing it well; the patterns (IX) are the verdicts compressed into reusable shapes.

Heraclitus, from the epigraph page, gets the last technical word, because the whole discipline is an argument with him. No one steps in the same river twice — the world moves, the user changes, the corpus is edited under the cache's feet. The semantic cache is the engineering wager that he was only *mostly* right: that beneath the moving water there are stones — questions whose answers hold still long enough to be worth keeping — and that with enough geometry, enough judgment, enough suspicion, and enough arithmetic, a system can tell the stones from the current. When it can, answers arrive before the question finishes echoing. When it cannot, it must have the discipline to say so and step into the river again. That discipline — not the vector store, not the threshold, not the hit rate — is what this report was about.

The cheapest retrieval is the one that never runs. The costliest answer is the stale one served with confidence. Craft is knowing, per entry, per user, per moment, which of those two sentences is speaking.

Appendix: A Glossary of Caching, and Further Reading in Order

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A Glossary of Caching

TWENTY DEFINITIONS THAT, READ IN ORDER, RESTATE THE BOOK.

The last two definitions entered in the revision of mid-August 2026, with §; the count rose from eighteen with them.

Semantic cache. A lossy, approximate cache serving stored answers for new queries judged equivalent by embedding similarity. The only cache in the LLM stack that can be *wrong* — the property from which this entire report follows.

Prefix (prompt) caching. Provider-side, exact, token-level reuse of identical context prefixes at $\sim 90\%$ input discounts. Changes the cost of computing an answer; never avoids computing it.

KV-cache reuse. Research-line reuse of per-chunk key–value states beyond exact prefixes. Reuses computation, not answers; a different altitude, a different ledger.

The E -and- τ doctrine. The entire intelligence of a semantic cache lives in two objects: the encoder and the threshold. Promoted by this report: E trained for calibration, τ grown into a policy.

Concentration of measure. The high-dimensional condensation that makes thresholding sound: random directions are orthogonal by default; polar caps are exponentially starved; chance hits at taught thresholds are cosmologically impossible.

Polar bears. The cap's population: occupants placed by training, never by chance. Curated, not sampled — and therefore every cache failure is systematic, not unlucky.

Anisotropy. Real encoders occupy a cone, not the sphere; random texts score 0.2–0.4, not zero. Thresholds are dataset-relative folklore; the equator moves.

The overlap. The founding measurement of the modern literature: correct- and incorrect-hit similarity distributions overlap, differently per cached entry. The global threshold's death certificate.

Value of serving. $\text{VoS} = pg - (1 - p)L - \kappa$; serve iff positive, i.e. iff $p > L/(g + L)$. Third member of the house equation family; the $(1 - p)L$ term is the cache's signature.

Error contract. The operator states δ , the admissible fraction of wrong hits; per-entry learned curves keep it, exploration labels pay for it, and the demonstrated rate is reported. A threshold you cannot audit is a guess you automated.

Calibration (P-CHR). Threshold usability, not ranking quality: the precision-versus-hit-ratio curve. Rankers can excel and threshold uselessly; objectives, not scale, decide.

Five layers. What a RAG system can cache: answers, retrieved sets, intermediates, KV states, understanding. The taught singular “the cache” was layer one's portrait.

Evidence gauntlet. Freshness checked against provenance at serve time — overlap, version identity, lexical support — with clocks demoted to backstops. Invalidate by evidence, not by clock.

Unsafe-served rate. The fraction of hits whose answers current evidence no longer supports. The freshness metric that retired MRF.

Operational equivalence. Same tool, same effect — the relation agents need and cosine does not measure. For actions, equivalence is decided (canonical forms, state keys), never scored.

Key collision. An adversarial prompt tailored into a victim's cap — the bear suit. By theorem, locality (hit rate) and collision resistance are one quantity with opposite signs.

Scoped cache. Per-tenant, per-user keyspaces making leaks and collisions structural impossibilities. The only defense that does not bet; its absence, in 2026, is malpractice.

Break-even hit rate. $H^* = C_{\text{ops}} / N(C_{\text{full}} - C_{\text{hit}})$, with C_{full} taken after prefix discounts and H after scoping. Cheap models raise it; latency products ignore it; “do not build” is a verdict, not a defeat.

Governed layer. Cached understanding: comprehension performed once, at authoring time, under review, then served many times — OKF’s concept files, seated above layer five, keyed by concept rather than by query. Its expiry is an authored *absolute* date (*stale_after*), not a relative TTL: staleness as a plain date comparison, no reference to read time. Its cost is sustained human attention at the merge; the wager that agents change that economics is open.

Provenance laundering. Serving a cache hit stripped of the trust tier and attestation status of the knowledge that minted it, so an unverified draft emerges wearing the borrowed authority of instant delivery. Cure: the entry keeps its concept back-pointers; tier and lifecycle re-derived from the live bundle at serve time; expiry or deprecation of a cited concept invalidates by ID.

Further Reading, in Order

DEPENDENCY-ORDERED, NOT CHRONOLOGICAL: each item assumes the ones before it.

1. **Blum, Hopcroft, Kannan, *Foundations of Data Science*, Ch. 2** — the geometry: annulus, caps, the strangeness. Read before anything with a threshold in it.
2. **Dar et al., *Semantic Data Caching and Replacement* (VLDB 1996)** — the original meaning of the term: equivalence as a theorem. Read to feel what was traded away.
3. **Bang, *GPTCache* (NLP-OSS 2023)** — the modern reference architecture and, in hindsight, the strawman: one static threshold. Short; read as history.
4. **Gill et al., *Domain-Specific Embeddings for Semantic Caching* (arXiv:2504.02268)** — why the encoder is a specialist’s job; the synthetic-pair pipeline for the fine-tuning entry fee.
5. **Baral et al., *Closing the Calibration Gap* (arXiv:2606.19719)** — why the field’s offline metric lied; P-CHR and the objective-not-scale lesson. Read before trusting any sweep.

6. **Schroeder et al., *vCache* (ICLR 2026)** — the anchor: the overlap measurement, per-entry thresholds, the error contract. The report's Part III in primary-source form.
7. **Shah, *GroundedCache* (arXiv:2605.27494)** — freshness as an evidence gauntlet; the unsafe-served rate. The layer-one answer cache made honest.
8. **Zhang et al., *Agentic Plan Caching* (NeurIPS 2025) and the *W5H2 canonicalization paper* (arXiv:2602.18922)** — the agent turn: cache structure, not answers; decide equivalence, don't score it.
9. **Zhang et al., *From Similarity to Vulnerability* (ICML 2026)** — the adversary: read last, so that every design you have just admired acquires its threat model — and its scoping requirement.
10. **McVeety and Hormati, "How the Open Knowledge Format can improve data sharing" (Google Cloud Blog, June 12, 2026); spec v0.2 at github.com/GoogleCloudPlatform/knowledge-catalog** — one addition from the revision of August 2026, placed after the adversary because the governed layer inherits him: understanding cached at authoring time, with papers — and §'s standing order that the papers travel through every cache built in front of it. OKFcRAFT is the shelf's full treatment.

Five things to remember, if only five

1. Only the semantic cache can be wrong; everything follows from that.
2. Chance never enters the cap: every hit is caused, every failure systematic.
3. The threshold is a decision with an error budget, not a number with a hope — and per entry, not global.
4. Scope is the only defense that does not bet; unscoped sharing is malpractice.
5. Prefix caching first, the scalpel second, and the break-even formula before either — "do not build" is a respectable answer.

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